

SIMATIC

Process Control System PCS 7 PowerControl Setup Guide

Function Manual

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Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

 DANGER
indicates that death or severe personal injury will result if proper precautions are not taken.
 WARNING
indicates that death or severe personal injury may result if proper precautions are not taken.
 CAUTION
indicates that minor personal injury can result if proper precautions are not taken.
NOTICE
indicates that property damage can result if proper precautions are not taken.

If more than one degree of danger is present, the warning notice representing the highest degree of danger will be used. A notice warning of injury to persons with a safety alert symbol may also include a warning relating to property damage.

Qualified Personnel

The product/system described in this documentation may be operated only by **personnel qualified** for the specific task in accordance with the relevant documentation, in particular its warning notices and safety instructions. Qualified personnel are those who, based on their training and experience, are capable of identifying risks and avoiding potential hazards when working with these products/systems.

Proper use of Siemens products

Note the following:

 WARNING
Siemens products may only be used for the applications described in the catalog and in the relevant technical documentation. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any problems. The permissible ambient conditions must be complied with. The information in the relevant documentation must be observed.

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Disclaimer of Liability

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

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Security information

Siemens provides products and solutions with industrial security functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial security concept. Siemens' products and solutions constitute one element of such a concept.

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial security measures that may be implemented, please visit

<https://www.siemens.com/industrialsecurity>.

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply the latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Security RSS Feed under

<https://www.siemens.com/industrialsecurity>.

Purpose of the Manual

This manual helps you to connect PCS 7 operator stations with PowerControl stations via PCS 7 PowerControl.

Aims

This manual provides you with the following information:

- Installing PCS 7 PowerControl
- Creating and configuring SIMATIC Manager projects
- Integrating PCS 7 PowerControl in IEC 61850 Intelligent Electronic Devices (IEDs)

It is assumed that you are thoroughly familiar with your programs and development environment.

Scope of This Manual

This manual applies to the following software versions:

- PCS 7 PowerControl V9.1

Structure of the Documentation

The PCS 7 PowerControl documentation includes the following:

- PCS 7 PowerControl Setup Guide (this document)
- PCS 7 PowerControl library objects manual
 - Describes how to use and configure technological and diagnostic objects from PowerControl
- PCS 7 PowerControl DBA manual
 - Describes the process and integration of technological and diagnostic objects of PowerControl into PCS 7 operator stations
- PCS 7 PowerControl Type Editor manual
 - Describes the process and expansion of PCS 7 PowerControl by adding customer-defined types
- Readme.txt files on the PCS 7 PowerControl installation media

Introduction and Installation

2.1 Security

2.1.1 Security concept

The security concept of SIMATIC PCS 7 should be considered a set of recommendations and is intended to support SIMATIC customers in creating a network for production plants fulfilling higher security requirements. The recommendations are based on the latest technology, current standards and the features of the employed products.

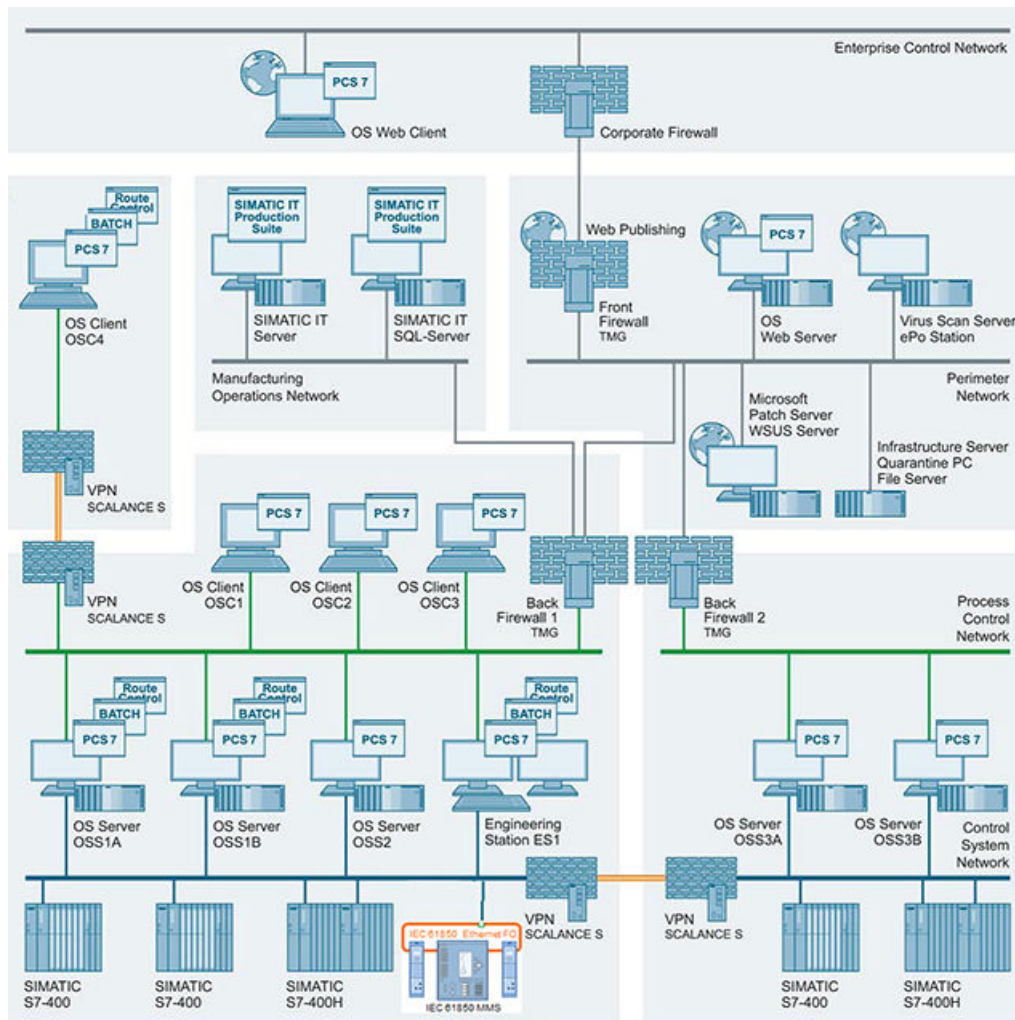
Because the threat to security is constantly changing, even a complete implementation of the security concept cannot permanently guarantee complete and one hundred percent protection. We therefore recommend a regular evaluation of the measures implemented as part of the security concept.

PCS 7 Security Concept Recommendations and Notes:

<http://support.automation.siemens.com/WW/view/en/22229786> (<http://support.automation.siemens.com/WW/view/en/22229786>)

2.1.2 PCS 7/PowerControl example

2.1 Security



2.1.3 General security information

Siemens offers IT security mechanisms for its automation and drive product portfolio in order to support the safe operation of the plant and machine. Moreover, our products are continuously developed further with regard to IT security. We therefore recommend that you keep informed about updates and upgrades for our products and always use the most recent version of each product. For more information, go to:

<http://support.automation.siemens.com> (<http://support.automation.siemens.com>)

You can also register for a product-specific newsletter here.

For the safe operation of a plant or machine, however, it is also necessary to integrate the automation components into an overall IT security concept for the entire plant or machine which corresponds to state-of-the-art IT technology.

For more information, go to:

<http://www.siemens.com/industrialsecurity> (<http://www.siemens.com/industrialsecurity>)

Products of other manufacturers should also be taken into account.

The following section provides a list of security aspects to which particular attention should be paid when installing and using PCS 7/PowerControl:

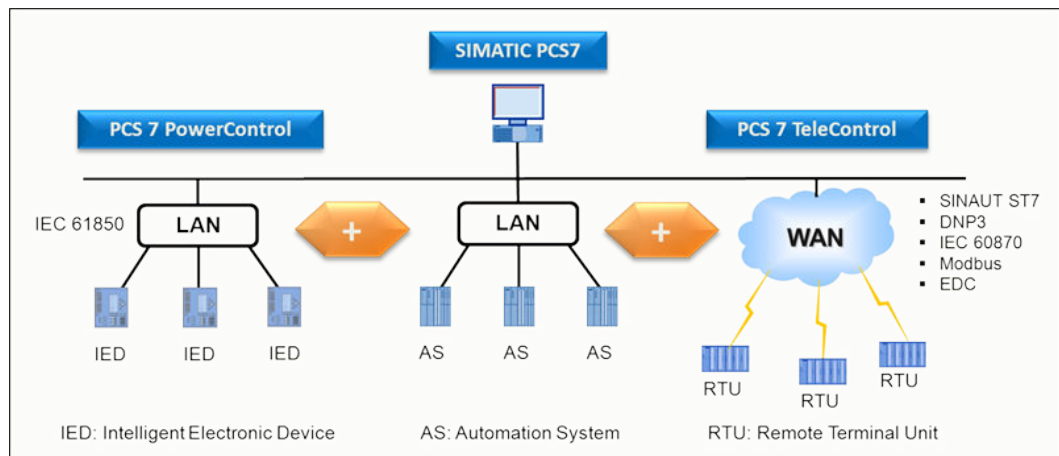
- Ensure the integrity of the data exchanged between the systems. It is particularly important to taken adequate measures to ensure that the data in these files cannot be manipulated. This applies, for example, to files of other systems that can be read by PCS 7/PowerControl, such as CSV files with configuration data. Other examples are XML, CSV and ZIP files that are supplied by the PCS 7/PowerControl system, such as DBA archive files and exports from DBA. Consider using a password-protected container for the exchange of such files.
- Do not change the PCS 7/PowerControl installation/upgrade files. For correct installation or correct upgrade of the PCS 7/PowerControl system with the required integrated security, it is important that these files are not changed. Therefore, the customer is responsible for ensuring that the person in charge of this process understands this measure. In some exceptional cases, the documentation describes that some files can be changed manually. The instructions for manual changes must be adhered to strictly. No other changes are permitted. If you have problems with specific files, please contact Siemens Support.
- The engineering and runtime components of PCS 7/PowerControl only function if the user currently logged on is configured as a member of the "SIMATIC HMI" user group. For data files that access these components we recommend that the folders containing such files are also configured with similar access rights to ensure that these files are not manipulated. Examples are the .DBA project file (Engineering) and WinCC project folders including all subfolders (Runtime).
- The disposal of PCS 7 PowerControl systems and components should be done following the measures as described in the *Process Control System PCS 7 Compendium Part F - Industrial Security Configuration Manual*, chapter 11 (Disposal of systems and components).

2.2 Overview

Introduction

Historically there has been a strict separation between Process Automation systems and Power Automation Systems. PCS 7 PowerControl combines Process Automation and Power Automation within one system.

PCS 7 PowerControl is a set of applications to allow the integration of intelligent electronic devices (IEDs) (e.g. SIPROTEC protection devices) into a PCS 7 automation system.

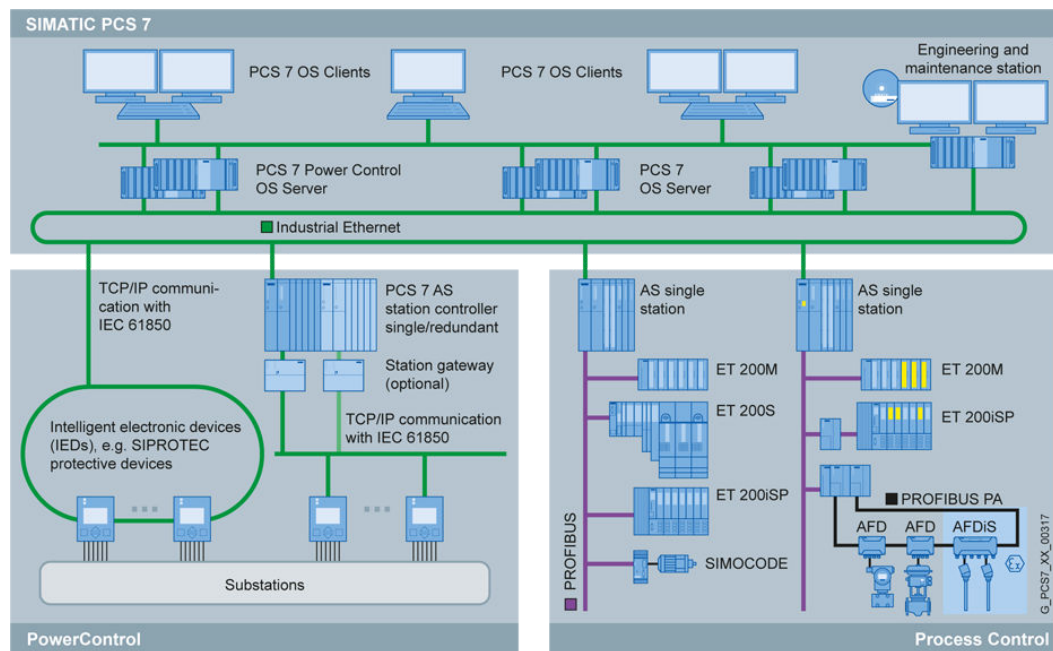


PCS 7 PowerControl can be used on:

- A PCS 7 Single Station system
- A single or redundant PCS 7 OS Server
- A single or redundant PCS 7 OS Server that also connects to PCS 7 S7-400s

PCS 7 PowerControl supports the IEC 61850 communication protocol. The PCS 7 PowerControl server includes a new driver (channel), which is acting as IEC 61850 client for the IEC 61850 server devices (IEDs). The communication channel is using MMS (Manufacturing Message Specification) based on TCP/IP.

The picture below shows a configuration with redundant PCS 7 OS Servers.



Benefits

The integration of Process Automation and Power Automation into one system offers the following benefits:

- Common look and feel for graphical elements (usage of PCS 7 APL style symbols and faceplates) leads to reduced risk of operation failures
- Support of plant hierarchy for picture navigation and group alarms
- Common operator philosophy (e.g. for alarms)
- Less engineering- & maintenance effort
- Uniform software platform
- Powerful BULK-Engineering (for 3rd party IEDs as well)
- due to high standardization level of IEC 61850 PCS 7 PowerControl can integrate all IED's which conform to the standard
- Siemens provides a large variety of IEDs as devices of the SIPROTEC product line

Processing Functions

In many cases, IEDs provide limited processing capability compared to the functions of a DCS controller. PCS 7 PowerControl adds the capability to incorporate Visual Basic Scripts that can perform preprocessing of IED raw data.

PCS 7 PowerControl Library contains symbols and faceplates for the SIPROTEC family of IED's. (Refer to the PCS 7/PowerControl Library Objects Guide for details).

Diagnostic Objects

PCS 7 PowerControl offers various diagnostic functions available for each IED:

- Main and backup connection state
- General request state
- IED time sync state
- Message counters
 - Successful transmissions
 - Unsuccessful transmissions
 - Retries

On connection failure:

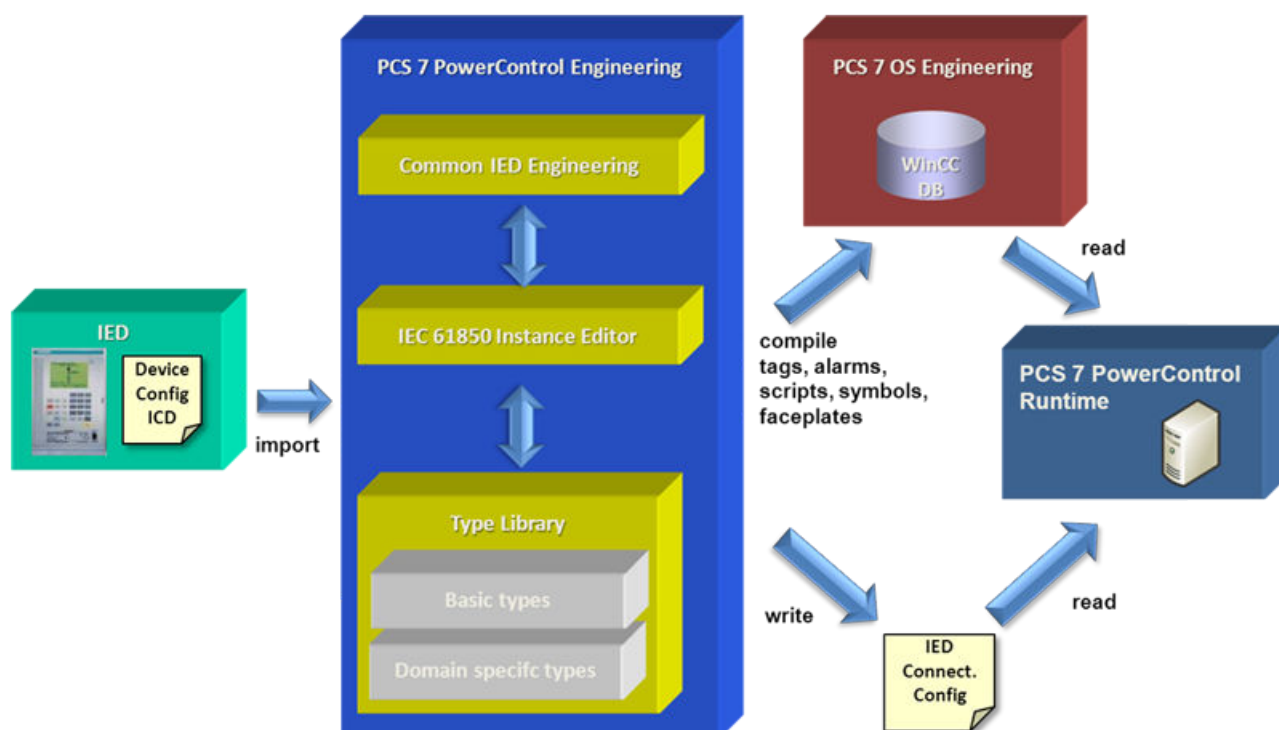
- All process objects of affected IEDs are marked as invalid
- Commands are blocked

PCS 7 PowerControl Engineering

The PCS 7 PowerControl engineering offers the following features:

- Object orientated (type/instance concept)
- OS Object library with function blocks , symbols and faceplates
- Automatic generation of OS object data (tags, alarms, graphical elements)
- Integration of additional IEC 61850 devices by importing device description files
- (ICD files) containing information regarding object types and instances
- Matching algorithm checks imported objects, whether they can be mapped to already existing PCS 7 PowerControl object types
- PCS 7 PowerControl instance editor enables creation of new additional data objects if no match / partial match of imported objects
- IEC-61850 IED cloning offers creation of complete replica of the IED, including all object instances, attribute configuration and plant view assignment

The following picture demonstrates the overall engineering workflow:



2.3 Software installation - Overview

PCS 7 PowerControl can be installed as a PCS 7 Single Station or in single or redundant client server architectures.

The table below describes the various components that need to be installed. Depending on the architecture of your system, these components may need to be installed on the different computer nodes in the architecture.

Note that all steps might not be required:

- If your computers have been pre-installed with the correct version of the Operating System, the installation of the Operating System is not required and Step 1 can be skipped.
- If you require no third-party software, step 2 can be skipped.

Step	Procedure
1	Installing the operating system (Page 14)
2	Installing Third Party Software (Page 15)
3	Installing PCS 7 software (Page 15)
4	Installing PCS 7/PowerControl Software (Page 15)
5	Install Authorizations and Licenses (Page 25)

Note

If you are upgrading an existing project, ensure that you have archived your project and saved the archive to media or another computer before installing software on your stations.

Note

Administrator rights are required to install the software. In addition, disable any virus protection software that is currently running.

Note

All computers on the network must have the same user accounts and passwords.

2.3.1 Installing the operating system

The following operating systems are supported by PCS 7/PowerControl:

- Microsoft Windows 2019 LTSC
- Microsoft Windows Server 2019

You can find detailed instructions on installing the operating system here:

- pcs7-readme.mht file in the root directory of PCS 7 installation DVD1 of 2 (PCS7 V9.0)
- SIMATIC PCS 7 DVD 1/2 in the folder _Manuals\English "PCS 7 - PC configuration.pdf"

2.3.2 Installing Third Party Software

Third party software (e.g. Microsoft Office, PC Anywhere, Virus software, etc.) should be installed prior to installing PCS 7. Refer to the PCS 7 Readme and PC Configuration and Authorization .PDF file for further information.

Any third-party Engineering Tools (e.g. Modbus, IEC, DNP3) should also be installed at this point.

2.3.3 Installing PCS 7 software

For detailed installation instructions, refer to:

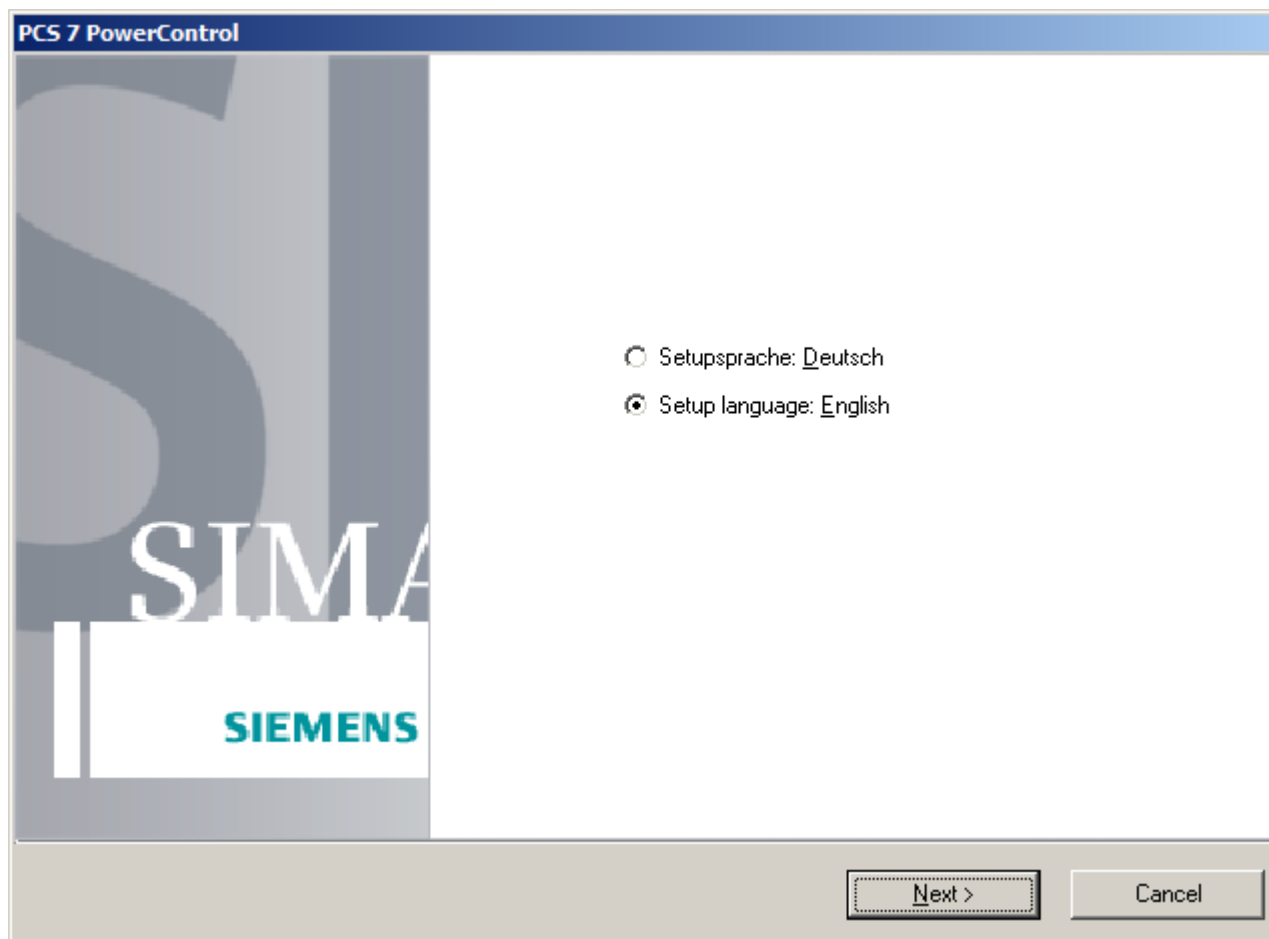
- pcs7-readme.mht file in the root directory of PCS 7 USB flash drive
- SIMATIC PCS 7 USB in the folder _Manuals\English "PCS 7 - PC configuration.pdf"

2.3.4 Installing PCS 7/PowerControl Software

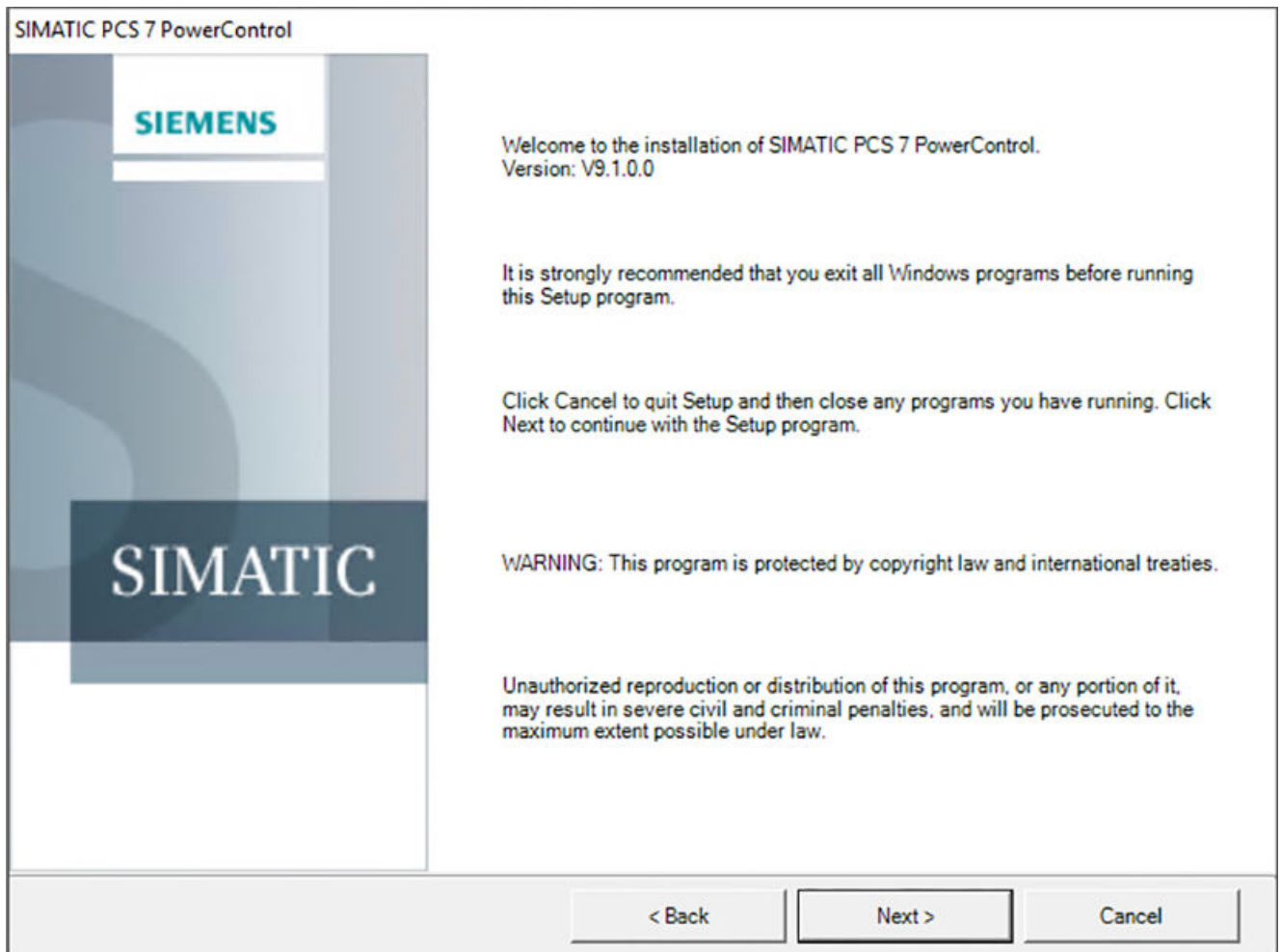
Note

Before installing this software you must log in with administrator rights and deactivate all anti-virus software that may be running at that time. Furthermore, this software should only be installed after all other PCS 7 software. Make sure that all login activity for PCS 7 is completed. You can find further information in the PowerControl file "Readme.wri" on the USB flash drive.

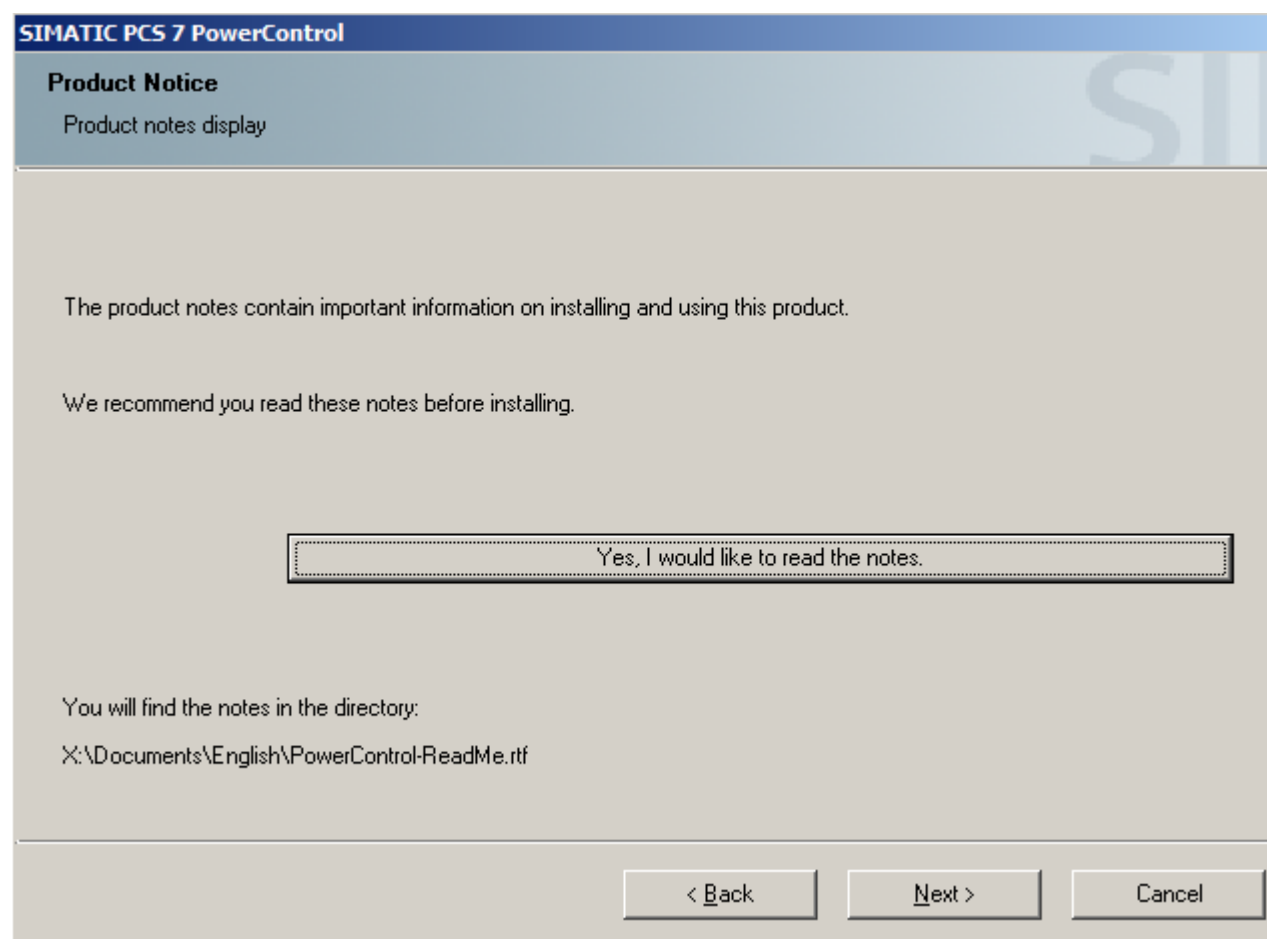
1. Place the PCS 7 PowerControl CD into the CD-ROM drive.
The setup program starts automatically. A dialog box for selecting the language opens.



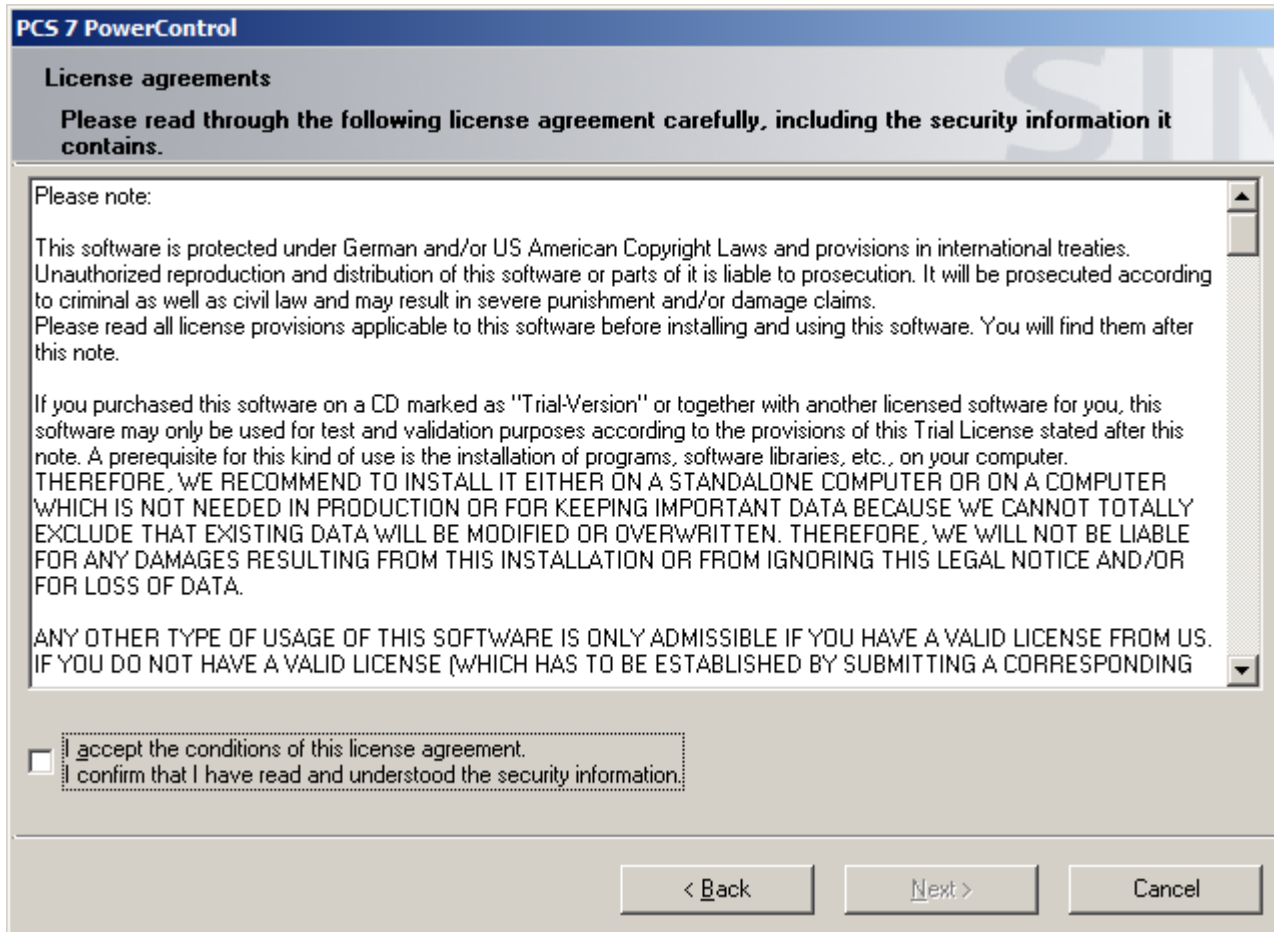
Choose the desired installation language (English or German).
The **welcome screen** opens.



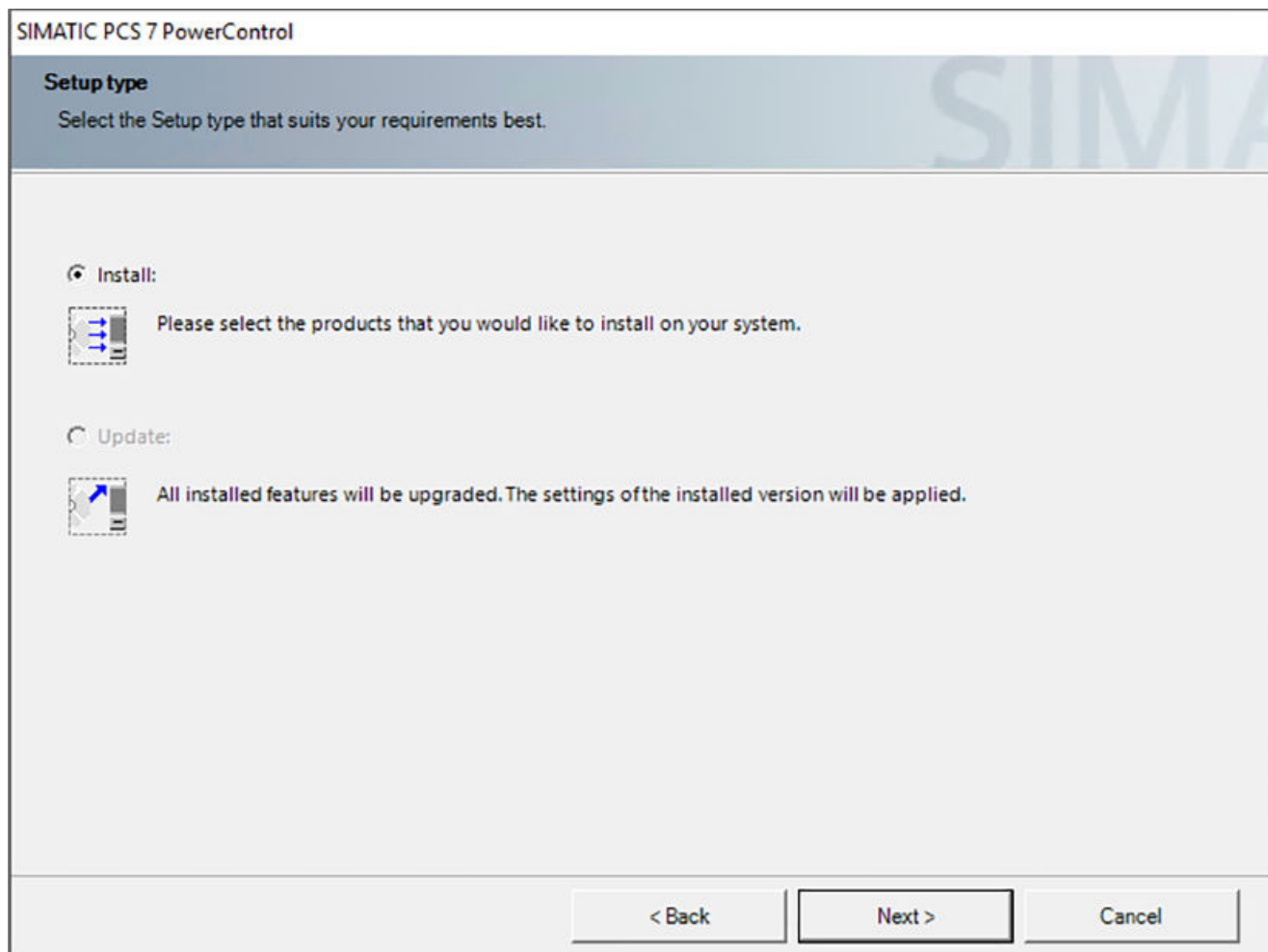
2. Click the **Next** button to continue. The **Product Notice** dialog opens.



3. To display the release information click on the button with the text **Yes, I would like to read the notes.**
Click the **Next** button to continue.
The **License agreements** dialog box opens.



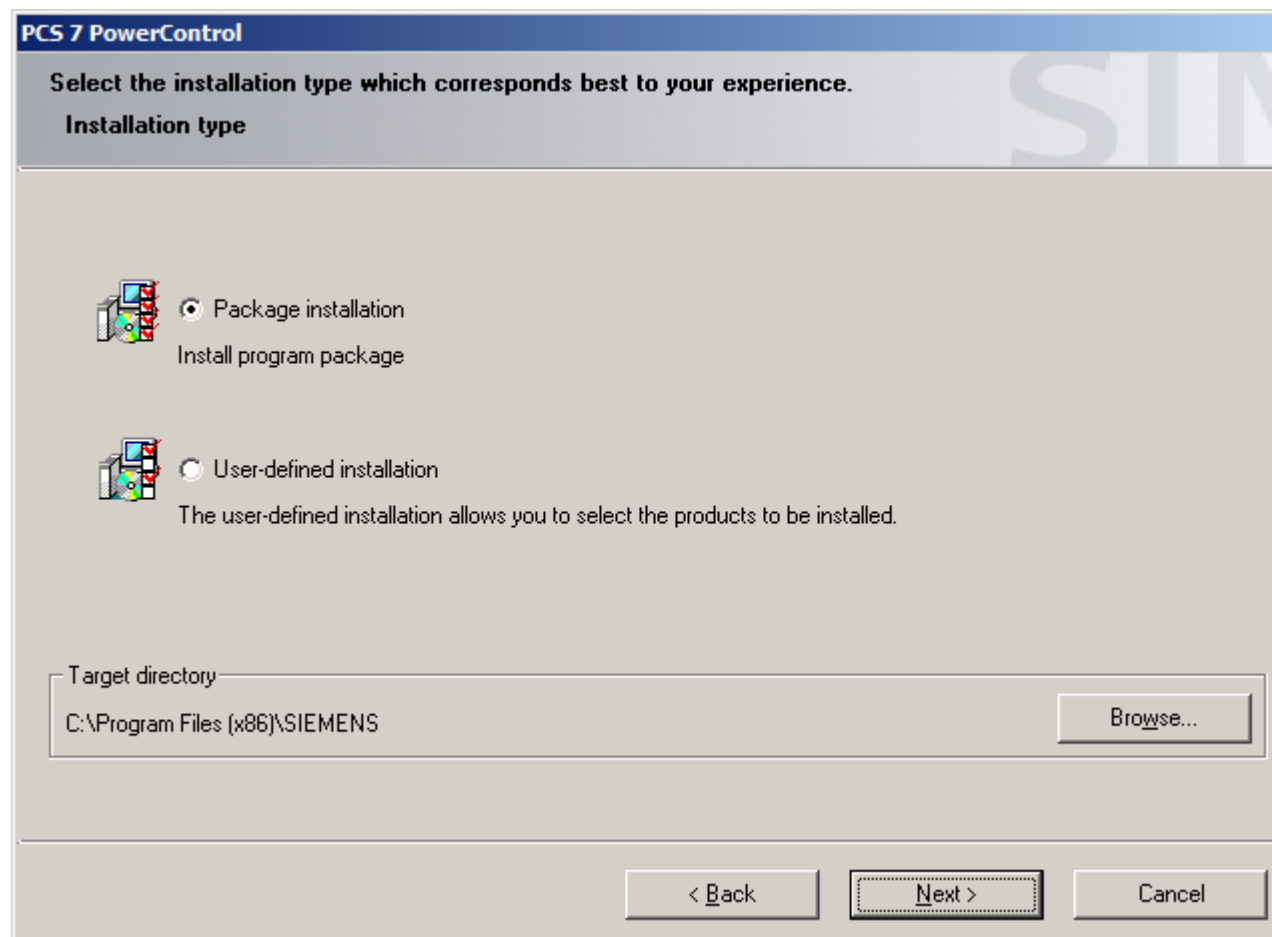
4. Read the license agreement.
Choose the option button **I accept the conditions of this license agreement**.
Click the **Next** button to continue.
The **Setup type** window opens.



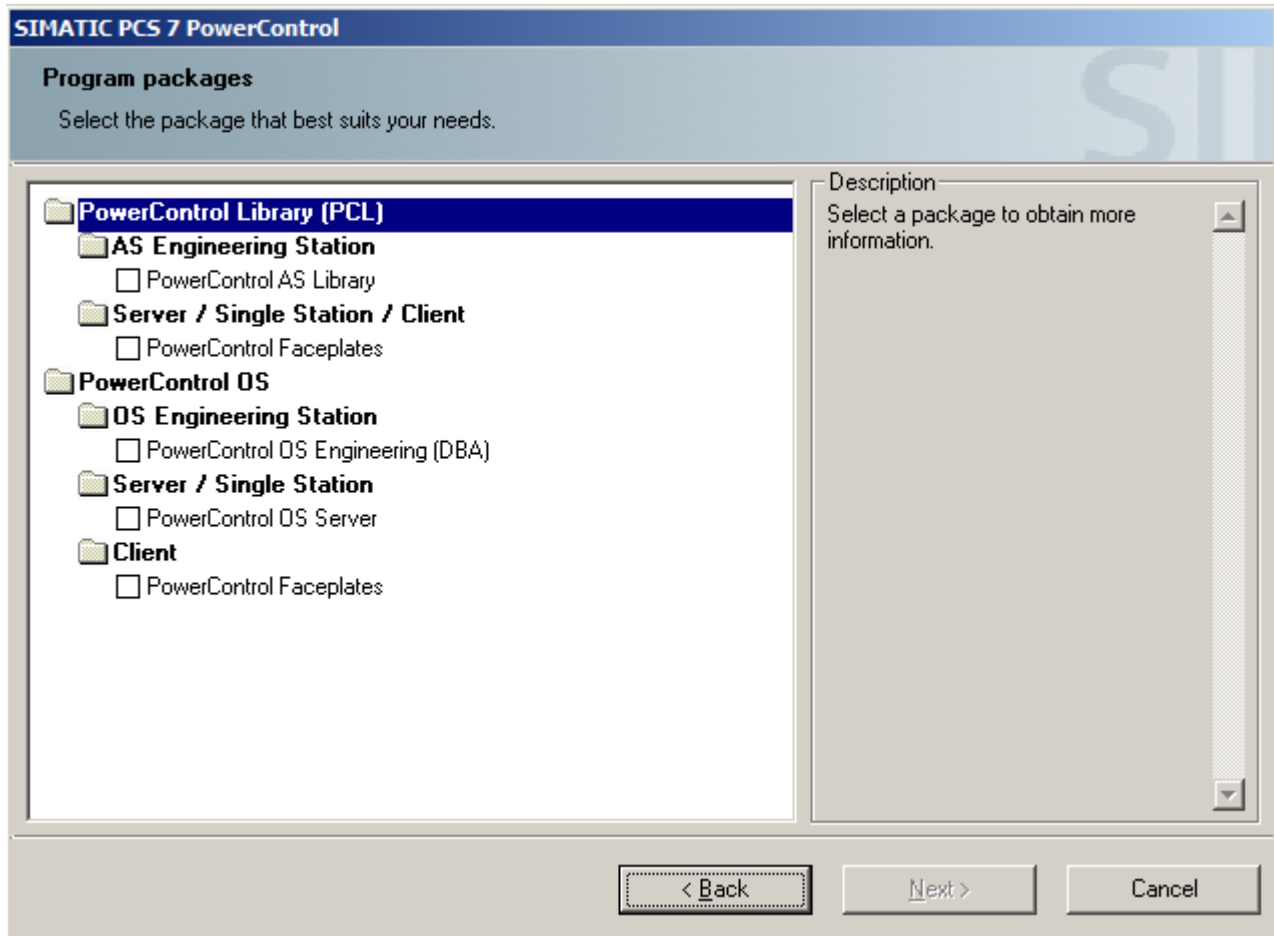
5. Select the setup type. Click the **Continue** button to progress.
The **User information** dialog box opens.

The screenshot shows a Windows-style dialog box titled "PCS 7 PowerControl". The main heading is "User information" with a sub-instruction "Please enter the required information." Below this, there are two text input fields. The first is labeled "Name:" and contains the text "Windows User". The second is labeled "Company:" and is currently empty. At the bottom of the dialog, there are three buttons: "< Back", "Next >" (which is highlighted with a dashed border), and "Cancel".

6. Enter the **username** and **organization**. Click the **Continue** button to progress.
The **Installation type** dialog box opens.



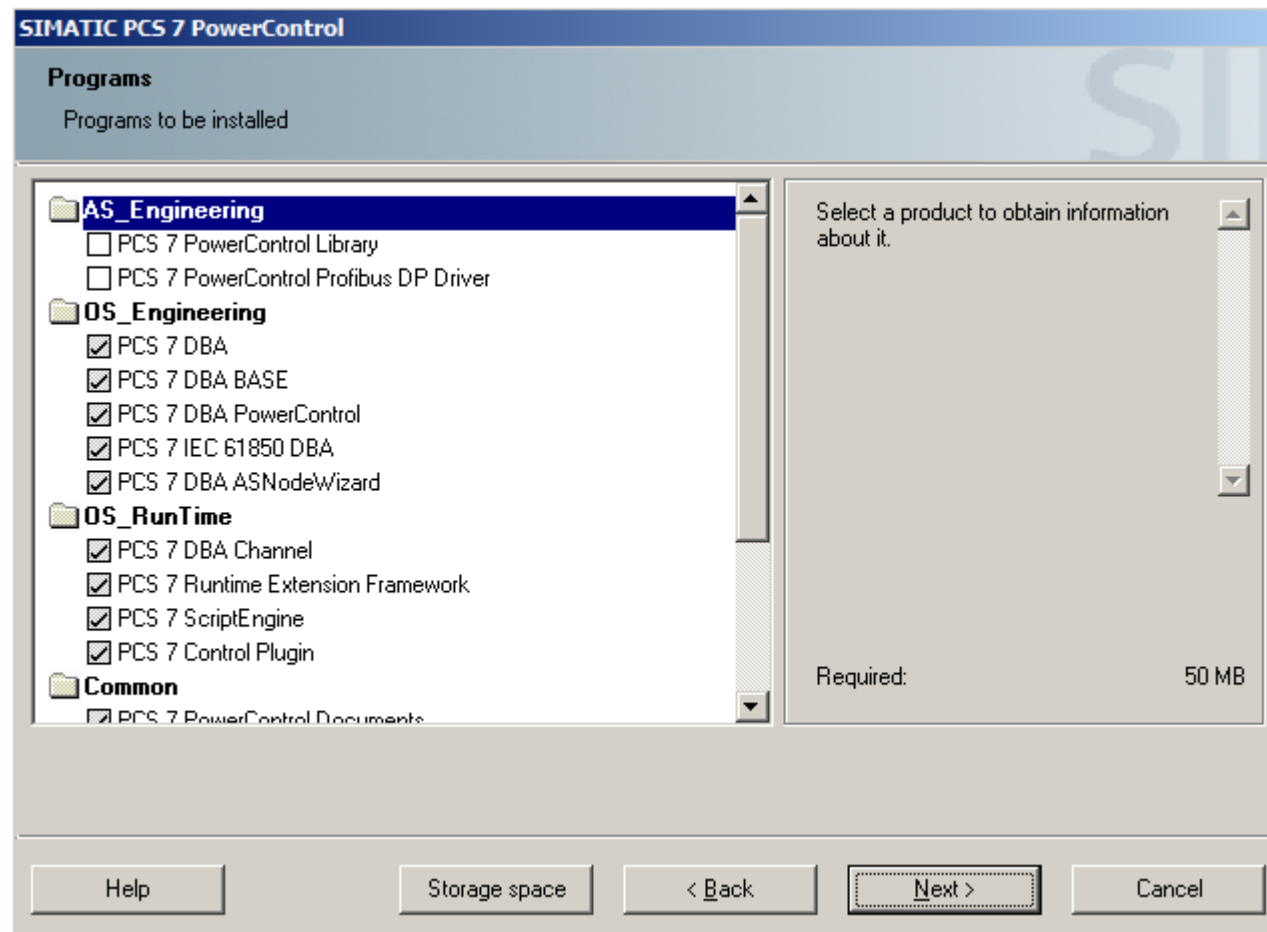
7. Select the installation type and click on the **Next** button. When the **Package Installation** is selected, the **Program packages** dialog box opens. When **User-Defined installation** is selected, the installation program continues directly with the "Programs" dialog box (step 8).



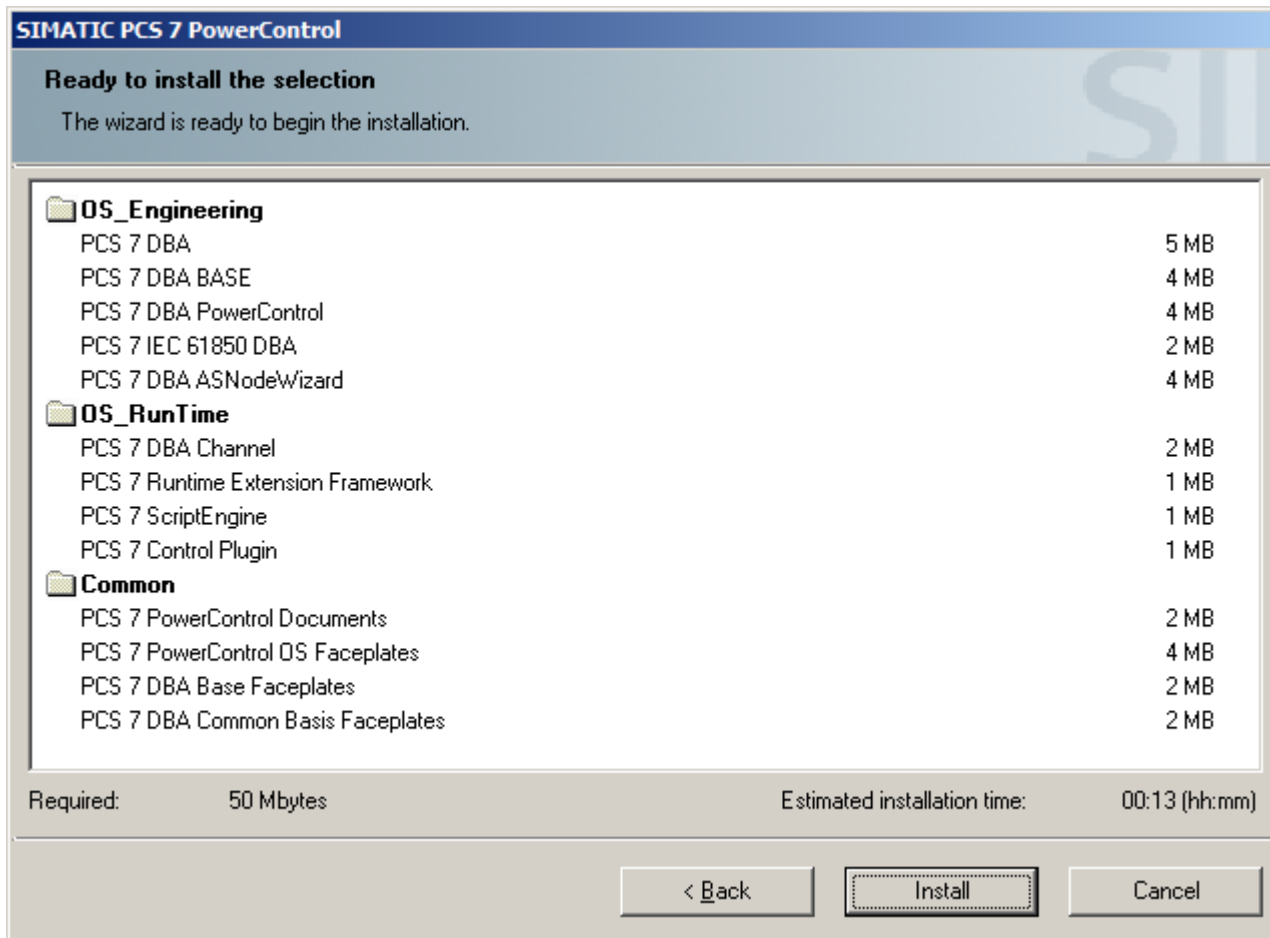
The top-level package PowerControl Library (PCL) applies to use cases 3, 4, and 5 as described in Use cases (Page 27). Select the relevant subpackage based on the type of station on which the package should be installed.

The top-level package PowerControl OS applies to use case 1 as described in Use cases (Page 27). Select the relevant subpackage based on the type of station on which the package should be installed.

8. Click on the relevant checkbox to select the package appropriate for the computer on which this installation is to be performed.



9. You can find the right components for the installation on a station in the table in PCS 7 PowerControl . The selection depends on the station type (e.g. client or server) on which to be installed.
Click on **Next** to adopt the programs selected.
The **Ready to Install the Selection** dialog box opens.



10. Click **Install** to start the installation process. The package installation begins.
The installation is performed through to completion. At the end of the installation you may be asked to restart the computer.

2.3.5 Install Authorizations and Licenses

Note

Each PCS 7/PowerControl Software package may include a License Key memory stick. As you install the various packages, be sure to install the license disk with that package.

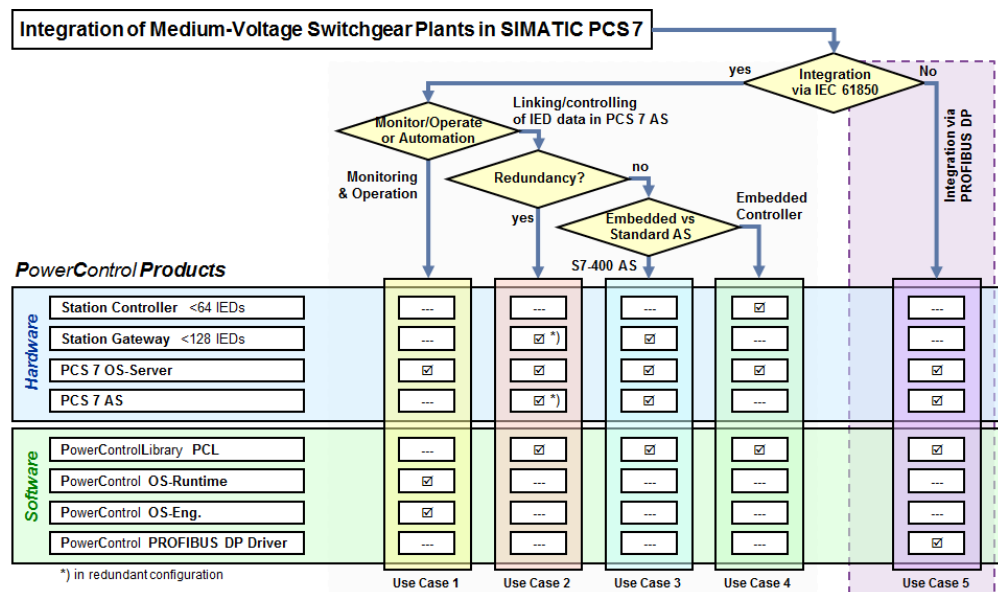
1. Insert the License Key memory stick that came with your software into USB port.
2. From the Windows taskbar, click **Start button > SIMATIC > License Management > Automation License Manager**.
3. Double-click the USB drive containing the License Key memory stick.
4. Select all of the authorizations.
5. Transfer the authorizations from the USB drive to the target drive.
6. Remove the License Key memory stick from the USB port.

2.4 Use cases

2.4.1 Use cases

Use cases

We can basically distinguish between five use cases. Cases 1 to 4 are based on the industrial standard IEC 61850, whereas case 5 describes the classic integration of protection devices via PROFIBUS DP.



Use case 1

IEC 61850 IEDs connected directly to the SIMATIC PCS 7 plant bus are only operated and monitored by the PCS 7 operator station. There are no connections to functions in a PCS 7 automation system, e.g. interlocks.

Documents:

- PCS 7 PowerControl DBA User Manual
- PCS 7 PowerControl DBA Type Editor
- PCS 7 PowerControl Library Objects
- PCS 7 PowerControl Setup Guide
- PCS 7 Technological Objects for PowerControl

2.4 Use cases

Use case 2

Use of IEC 61850 IEDs which are connected to a PCS 7 automation system, e.g. for interlocks. Due to higher availability requirements, the automation system and the Station Gateway are redundantly configured.

Documents:

- PowerControl Library AS Manual

Use case 3

Use of IEC 61850 IEDs which are connected to a PCS 7 automation system, e.g. for interlocks. The automation system is a modular PCS 7 standard controller together with a Station Gateway.

Documents:

- PowerControl Library AS Manual

Use case 4

Use of IEC 61850 IEDs which are connected to a PCS 7 automation system, e.g. for interlocks. The automation system is designed in the form of a station controller.

Documents:

- PowerControl Library AS Manual

Use case 5

Use of SIPROTEC protection devices with PROFIBUS DP interface which are connected to a PCS 7 automation system via PROFIBUS.

Documents:

- PROFIBUS DP Driver Blocks for PowerControl

Additional information:

https://www.automation.siemens.com/w2/efiles/pcs7/pdf/00/br_powercontrol_2013_de_Web.pdf

Project Engineering Overview

Engineering a PowerControl project can be divided into the following high level tasks:

Task	Description	Section
1	Creating a PCS 7 PowerControl Project (Page 30)	3.1
2	AS/IED Configuration and Programming (Page 31)	3.2
3	Integrating PCS 7 PowerControl with IEC 61850 Intelligent Automation Devices (IEDs) (Page 32)	3.3
4	Configuring PCS 7 PowerControl OS Stations (Page 33)	3.4

3.1 Creating a PCS 7 PowerControl Project

A PCS 7 PowerControl project is created using SIMATIC Manager. The process is described in detail in Creating a SIMATIC Manager Project (Page 37). This includes descriptions of how to configure all of the PC Stations in your system. Once these steps have been completed, the framework for additional engineering and maintaining your system is in place.

3.2 AS/IED Configuration and Programming

For configuring and programming the S7 family of controllers for Process Control applications, use SIMATIC CFC/SFC and STEP 7.

Configuring and programming objects and object behavior in IEC 61850 IEDs is beyond the scope of this guide. For configuring SIPROTEC IED devices, use DIGSI 4. For configuring and programming third party IEC 61850 IED's use the programming tool(s) provided with the hardware.

3.3 Integrating PCS 7 PowerControl with IEC 61850 Intelligent Automation Devices (IEDs)

PCS 7 PowerControl can be integrated with Intelligent Automation Devices (IEDs).

In order to integrate an IED it is necessary to have access to an ICD (IED Capability Description) or SCD (System Capability Description) file that defines the device using the IEC 61850 Substation Configuration Language (SCL). In addition, the following parameters about the device must be known (for some devices this information is stored within the ICD/SCD file and will be extracted automatically as needed by PCS 7/PowerControl:

- TCP/IP Address
- OSI Parameters
- Authorization information (if authorization is enabled for the device)

See also

Creating a SIMATIC Manager Project (Page 37)

3.4 Configuring PCS 7 PowerControl OS Stations

Once IED objects have been created and programmed, the objects must be made accessible in PCS 7 PowerControl OS stations.

To do this, a mapping tool is provided, PCS 7 PowerControl DBA, that simplifies this process.

DBA provides:

- Automatic generation of the PCS 7 PowerControl OS database including display hierarchy, needed variables, alarms, alarm-messages and priorities as well as function block symbols and faceplates.
- Automatic assignment of objects to the plant view and automatic placement of a symbol for the object on a picture.

The details of using DBA are provided in the *PCS 7 PowerControl DBA User's Guide* (Document Number A5E03987152-01).

This document includes step-by-step instructions for the basic workflow for defining and directing the mapping process.

3.4.1 Completing the PCS 7 OS Database Configuration

Once the PCS 7 OS database has been generated by DBA, the following additional configuration steps complete the PCS 7 OS database configuration.

3.4.1.1 Completing Graphic Displays

Based on the system hierarchy configured in DBA, and the placement of AS objects within this hierarchy, DBA automatically creates the dynamic portion of all graphic displays. Symbols for each object are automatically placed, in a tiled manner, in each picture.

To complete the graphic displays, create the appropriate static elements (pipes, tanks, etc.) and move the already-created dynamic symbols to the desired position on the graphic.

3.4.1.2 Station Lifebeat Monitoring Configuration

A Lifebeat Monitoring capability is provided that offers at-a-glance status information for all of the PC Stations, AS Stations, and AS connecting devices in your system. The basic variables needed to provide this capability are created by DBA in the PCS 7 OS database. An additional configuration step is required to complete the Lifebeat Monitoring configuration. Refer to Station Lifebeat Monitoring Setup (Page 67), for details regarding this configuration step.

3.4.1.3 User Administration

PCS 7 allows user rights to be established based on the layout of the system hierarchy. The highest level of this hierarchy is divided into Areas. This hierarchy, once configured in DBA, is automatically created in the PCS 7 database.

For more details on configuring Area-based User Administration, refer to the help file:

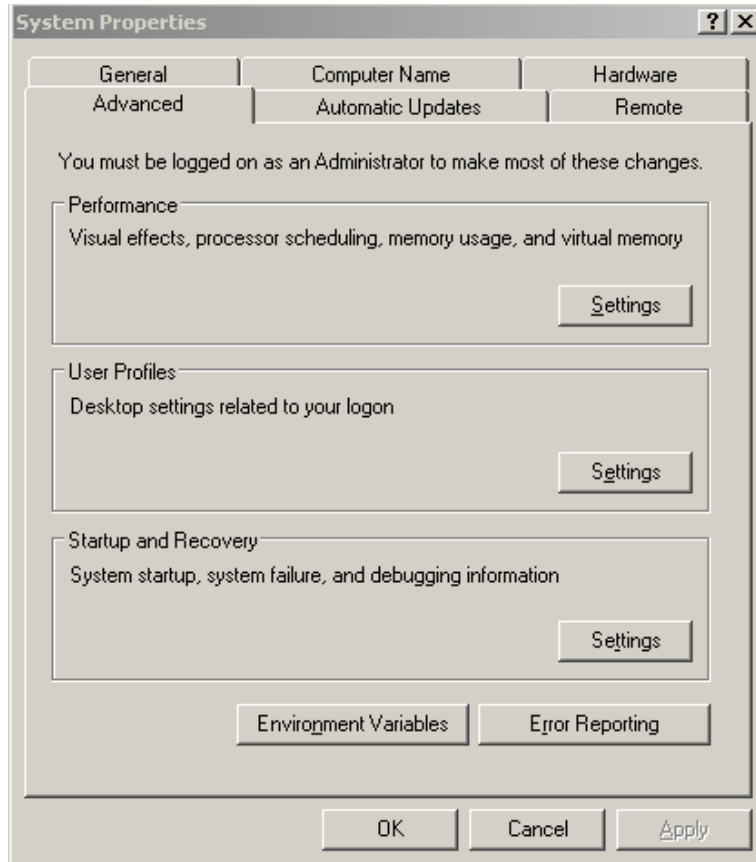
Start>Simatic>WinCC>WinCC Information System>Working With WinCC>Setting up User Administration

3.4.1.4 Special Instructions for Configuring a Backup OS Server

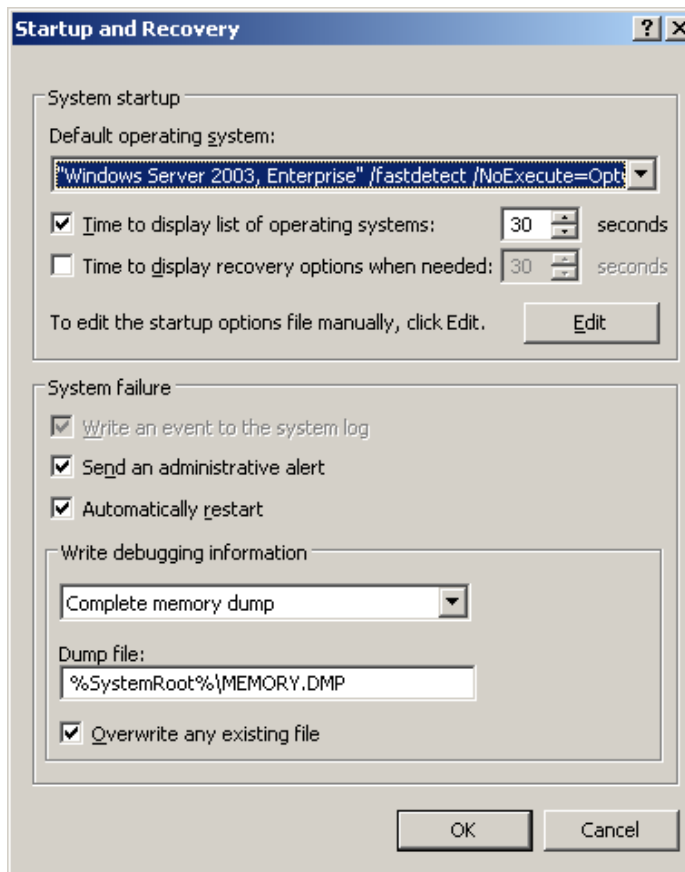
Special Instructions for Configuring a Backup OS Server

In redundant OS systems, an additional setup step is required to enable proper operation of archives.

1. From the Windows Control Panel, select System to invoke the System Properties dialog and select the **Advanced** tab.



2. Select the **Settings** button under Startup and Recovery.



3. Be sure that the "Time to display list of operating systems" entry is checked and use the listbox to select a value of 240 seconds.

This will ensure that when both servers are started up simultaneously, the standby server will not complete initialization until the master server is up and running. This is necessary to ensure that the archives are synchronized properly when both servers are started up simultaneously.

3.4.1.5 Downloading

In a client/server architecture, all PCS 7 OS databases (both client and server) need to be downloaded to the OS stations.

The process of downloading PCS 7 OS projects to PC Stations is described in detail in Chapter Downloading and Activating PCS 7 OS Stations (Page 73).

Creating a SIMATIC Manager Project

Note

On a Single Station, project engineering is done on the Single Station using SIMATIC Manager, and downloading to other stations is not required. In a single or redundant client server architecture, project engineering is done on the Engineering Station (ES). The project is then downloaded to all OS Server nodes and all OS clients from the ES.

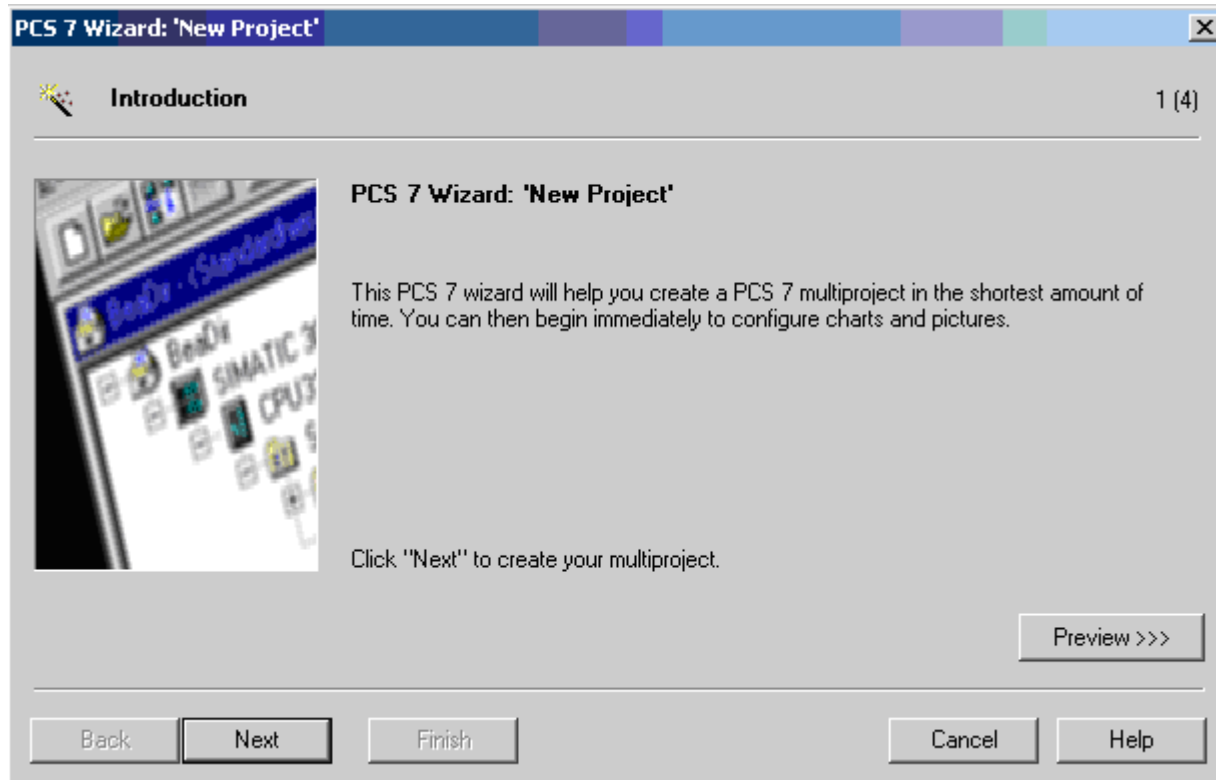
The project steps described below assume a single project with a redundant server pair and one client.

4.1 Create a New Project in SIMATIC Manager

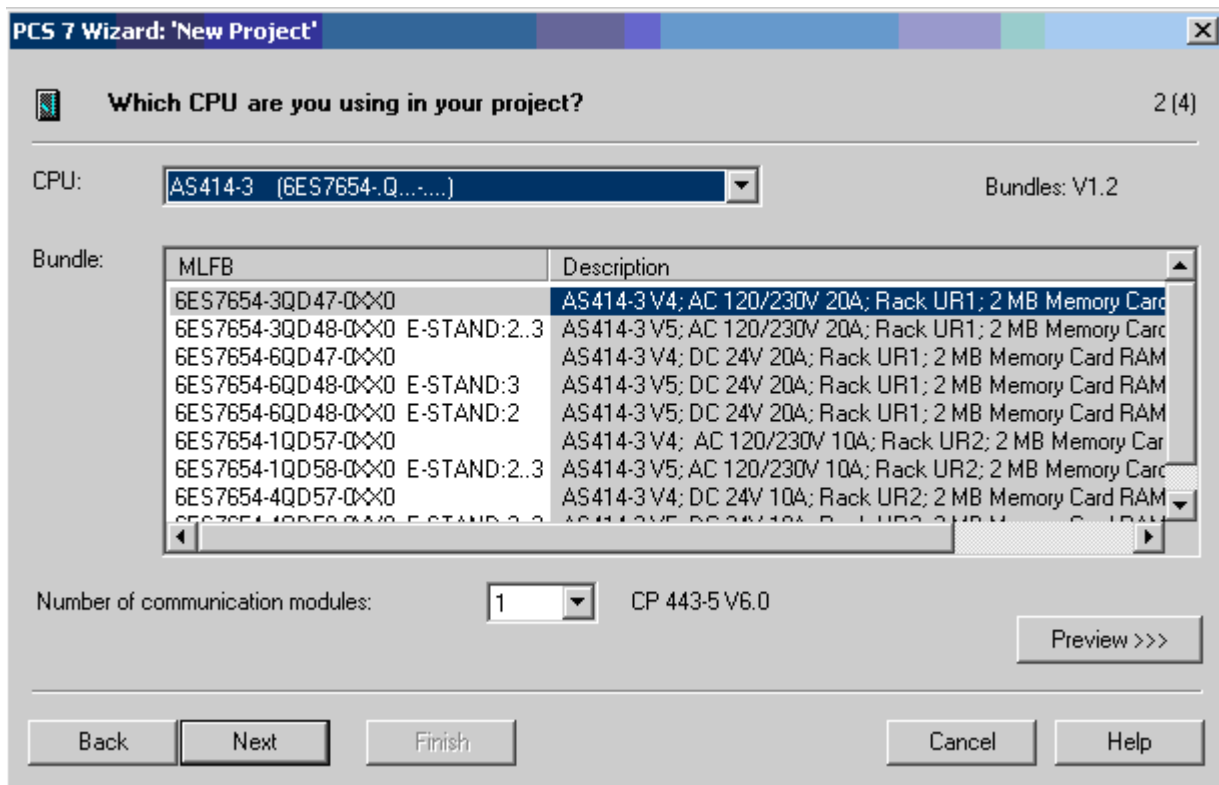
1. Open SIMATIC Manager, **Start > SIMATIC > SIMATIC Manager**
The PCS 7 Wizard 'New Project' dialog box opens.

Note

Press the **F1** function key for help on each screen of the PCS 7 New Project Wizard.



2. Click the **Next** button.
The **Which CPU are you using in your project?** dialog box opens.



The selections on this page are only relevant for dual channel projects (which include S7 controllers configured in Simatic Manager in addition to AS nodes configured with DBA). For projects that don't include S7 controllers, accept the default settings. For projects that do include S7 controllers refer to the S7 user documentation or press the Help button for more details.

4.1 Create a New Project in SIMATIC Manager

3. Click the **Next** button to accept the default.
The dialog box labeled **Which objects are you still using?** opens.

- In the **Plant hierarchy** area, the **Number of levels** should be configured when using S7-400 controllers in the same OS project as DBA-configured AS Nodes. It should be set to the same value used in DBA.
- In the **AS objects** area, clear **CFC chart** and **SFC chart** check boxes. The CFC and SFC selections on this page are only relevant for S7 projects. For S7 projects refer to the S7 user documentation or press the **Help** button for assistance.
- In the **OS objects** area, select the **PCS7 OS** check box, and the appropriate system type.
Single station system – a single PC station will be created with an OS project.
Multiple station system – two PC stations will be created, an OS server project, and an OS client project.
Multiple station system redundant – three PC stations will be created, two OS server projects and an OS client project.

Note

If your project requires additional server or client stations, they can be added once the wizard steps have completed. Refer to *Process Control System PCS 7 Engineering System Configuration Manual* for details regarding adding additional PC Stations.

4. Click the **Next** button.

The dialog box labeled **Where do you want to store the project? (or multiproject)** opens.

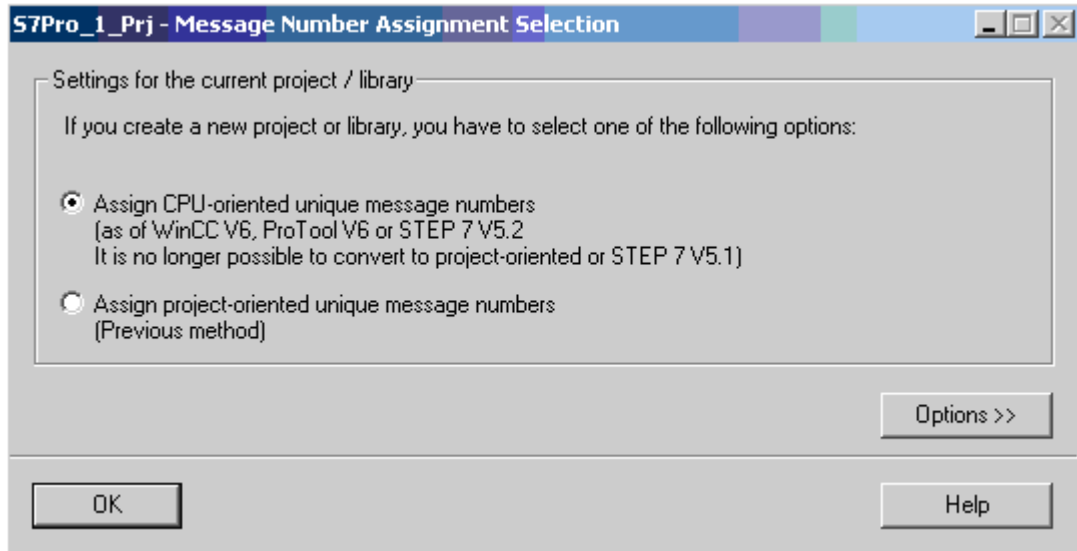
The screenshot shows the 'Where do you want to store the multiproject?' dialog box from the PCS 7 Wizard. The title bar reads 'PCS 7 Wizard: 'New Project''. The dialog has a tabbed interface with '4 (4)' tabs. The main area is divided into two columns. The left column contains a 'Directory name:' label with a text box containing 'S7Pro_1', a 'Storage location (path):' label with a text box containing 'C:\Program Files\SIEMENS\STEP7\s7proj', and a list of 'Available directories and files' including 'APACS_ALL', 'APACS_CS', 'APACS_SS', 'projD.pro', and 'TPN_KHE'. The right column contains the text 'The following objects will be created:' followed by a list: 'Multiproject: S7Pro_1_MP', 'Project: S7Pro_1_Prj', and 'Master data library: S7Pro_1_LIB'. There are 'Browse' and 'Preview >>>' buttons on the right. At the bottom, there are 'Back', 'Next', 'Finish', 'Cancel', and 'Help' buttons.

- In the **Directory name** box, enter a directory folder name or accept the default. This is the folder name for the directory under which the client/server projects will be created.
- In the **Storage location (path)** box, enter a path or accept the default. This path is where the directory folder will be located. This is the path for the master project on the engineering station.

4.1 Create a New Project in SIMATIC Manager

5. Click the **Finish** button.

The **Message Number Assignment Selection** dialog box opens.



6. Accept the default. Click the **OK** button.

The project is created. This setting applies to projects that include S7-400 controllers and does not apply to messages created by DBA.

4.2 Server Project Setup

The following steps are required to configure a PCS 7 OS Server project in SIMATIC Manager:

Step	Section
Create Shared Folders (Page 43)	4.2.1
Configure Primary Station (Page 44)	4.2.2
Configure Standby Station (Page 46)	4.2.3
Run OS Project Editor (Page 48)	4.2.4
Generate Server Data (Page 49)	4.2.5
Lifebeat Monitoring Setup (Page 50)	4.2.6
Configure Alarm Logging (Page 50)	4.2.7
Configure Tag Logging (Page 51)	4.2.8

Note

PCS 7 OS stations must also be configured under PC Stations in your corresponding DBA Project. Refer to Chapter 5 of the *PCS 7 PowerControl DBA User's Guide* for details concerning the configuration of PC Stations in DBA.

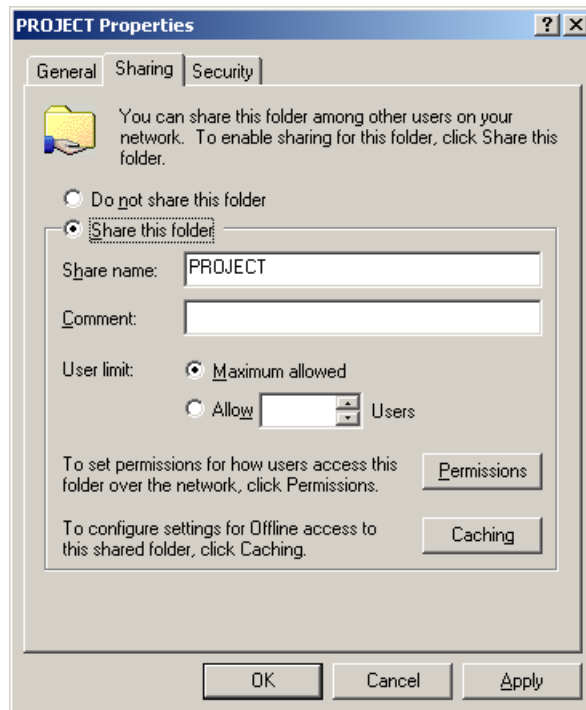
4.2.1 Create Shared Folders

Shared folders on the servers are needed to store the project created on the Engineering Station. When the project is downloaded, it will be transferred from the Engineering Station to these folders. You cannot configure your stations in SIMATIC Manager until these folders are created and properly set up. Follow these steps to create and set up these folders, if they don't already exist.

Shared folders must be created for both Primary and Standby Servers. Follow the steps in the following section to create a shared folder on each server.

Procedure for Creating a Shared Folder

1. Using Windows Explorer, create a new folder such as "PROJECT" on one of the station's drives. Projects created on the Engineering Station shall be downloaded to this folder on each node. If desired, you may create a "Projects" folder under **\\Program Files\\Siemens\\Step7\\s7proj** for your projects. However, the name of the folder and where you place it is up to you.
2. Set up sharing on the newly created PROJECT folder. Right-click the **PROJECT** folder and select **Sharing...** The **PROJECT Properties** dialog box opens.



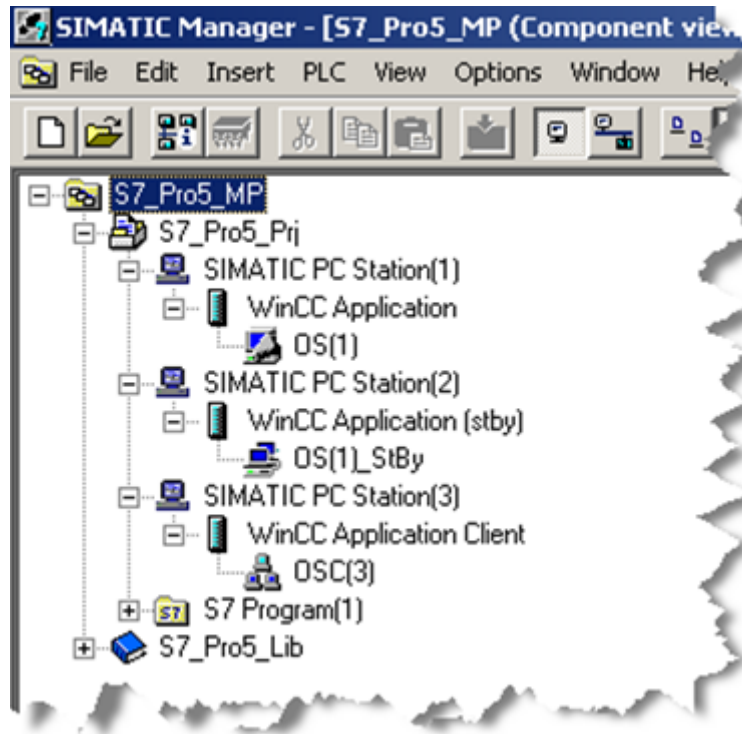
3. Select the **Sharing** tab, select the **Share this folder** radio button.
4. Click the **Permissions** button. The **Permissions** window opens. Select **Everyone**, and then click the **Full Control** check box in the **Allow** column.
5. Click **OK** to close the **Permissions** window.
6. Click the **OK** button to close the **Project Properties** dialog box.

4.2.2 Configure Primary Station

Identify the path to which the project will be downloaded on the Primary Server.

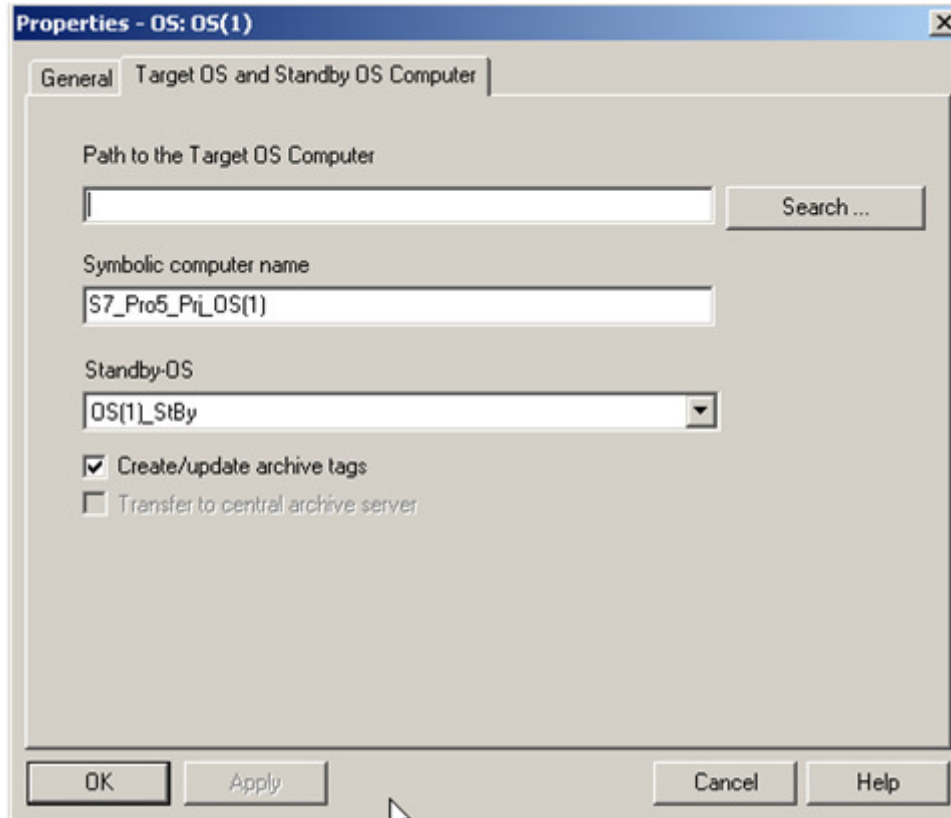
Follow these steps:

1. In the Component View of SIMATIC Manager, right-click the PC Station that is the Primary Server and select **Object Properties**. For example, if your Primary Server PC Station is SIMATIC PC Station(1) in the figure below, right-click on **SIMATIC PC Station(1)**.



2. In the **General** tab, enter the computer name of the primary computer in both the **Name** and **Computer name** fields. These names must be entered in ALL CAPS. Click the **OK** button.
3. Right-click **OS(1)**, and select **Object Properties**.
4. The **Properties** dialog box opens.

5. Rename, as desired, the **OS(1)** to the Server Project name.



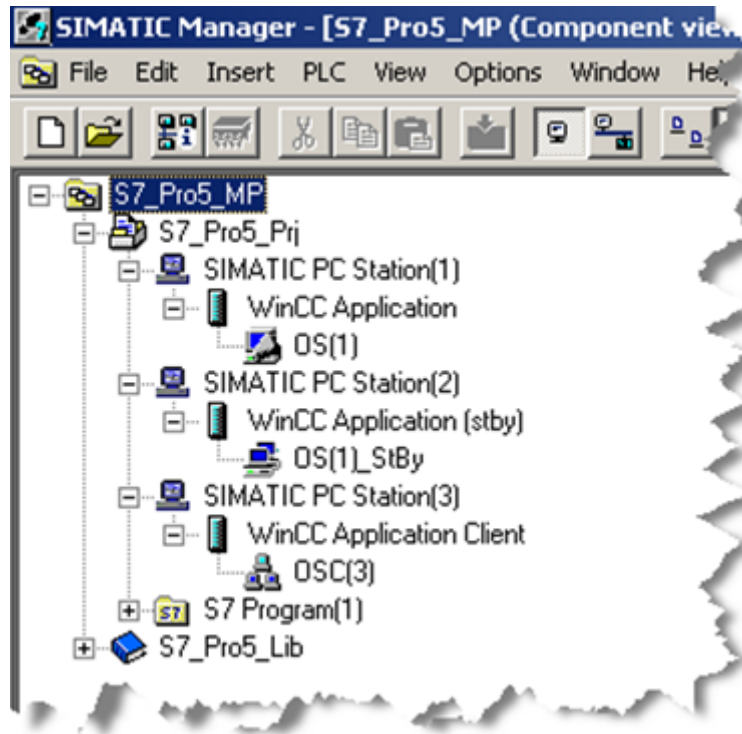
6. Click the **Target OS and Standby OS Computer** tab.
7. Select the Network button to browse and map to the shared project folder on OS Server. If a mapped drive already exists, select it from the available drives listed in the **Load In** list box. This is the path to the folder on your primary server where you will store and operate the project. This path must include the network name of the target computer.
Note: This folder must already exist on the primary and standby servers and be shared. This folder's permissions should be set to **Full Control** for **Everyone**. Refer to section **Create New Folders on the Primary and Standby Servers and All Clients** in this document.
8. Click the **OK** button.

4.2.3 Configure Standby Station

Identify the path to which the project will be downloaded on the Standby Server.

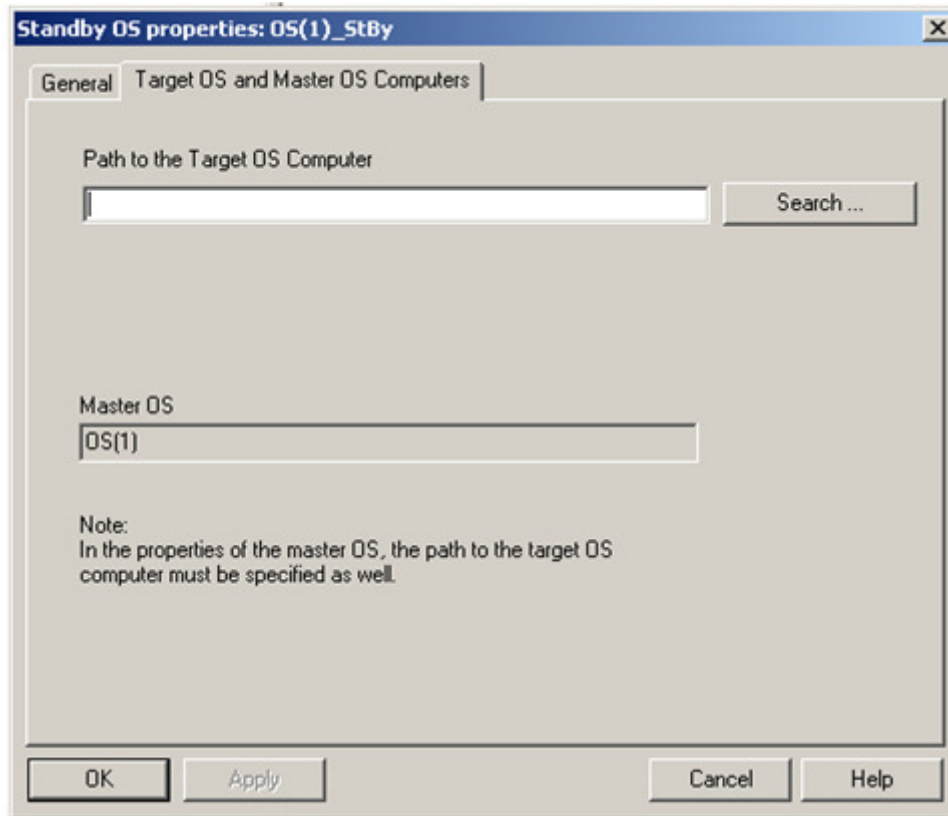
Follow these steps:

1. In the Component View of SIMATIC Manager, right-click the PC Station that is the Standby Server and select **Object Properties**. For example, if your Standby Server PC Station is SIMATIC PC Station(2) in the figure below, right-click on **SIMATIC PC Station(2)**.



2. In the **General** tab, enter the computer name of the standby computer in both the **Name** and **Computer name** fields. These names must be entered in ALL CAPS. Click the **OK** button.
3. Right-click **OS(1)_StBy**, and select **Object Properties**. The default project name of Primary Server/Project Name_StBy must be used. The **Properties** dialog box opens.

4. Click the **Target OS and Master OS Computers** tab.



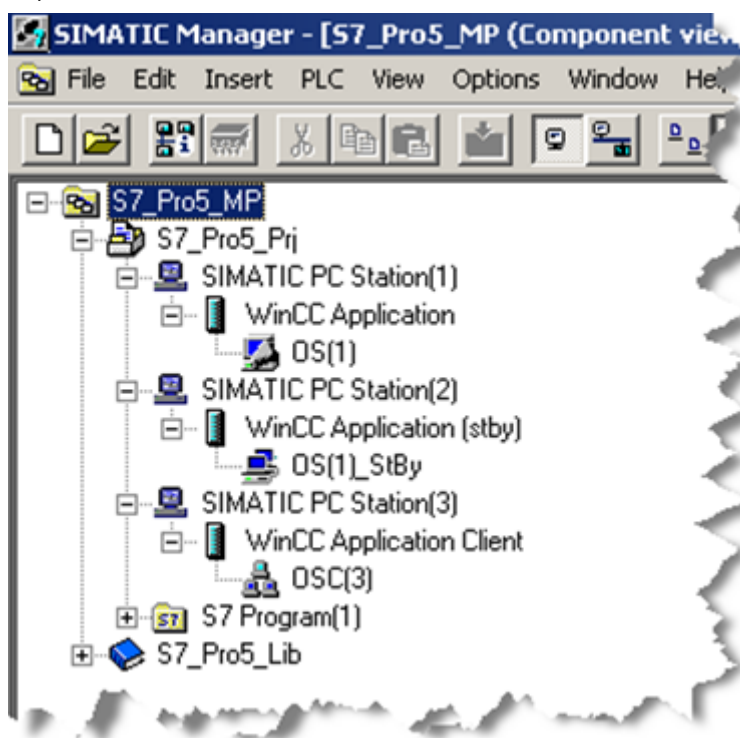
5. Select the Network button to browse and map to the shared project folder on OS Server. If a mapped drive already exists, select it from the available drives listed in the **Load In** list box. This is the path to the folder on your primary server where you will store and operate the project. This path must include the network name of the target computer.
Note: This folder must already exist on the primary and standby servers and be shared. This folder's permissions should be set to **Full Control** for **Everyone**. Refer to section **Create New Folders on the Primary and Standby Servers and All Clients** in this document.
6. Click the **OK** button.

4.2.4 Run OS Project Editor

From WinCC Explorer, run the OS Project editor in order to define the layout, screen resolution, monitor configuration, message configuration, message display, area, runtime window, and basic data. This step is also necessary in update your project to include all necessary library faceplate and symbols.

To run the OS Project Editor:

1. In the Component view of SIMATIC Manager, right-click the WinCC OS application and select **Open Object**. For example, if you want to run the OS Project Editor on the application OS(1) in the figure below, right-click on **OS(1)**. This will open the application using WinCC Explorer.



2. Select the OS Project Editor and make any desired changes. Refer to online help as you are running the OS Project Editor for more details.

4.2.5 Generate Server Data

1. From the Component view in SIMATIC Manager, right-click the primary OS server project, and select **Generate Server Data**.
When the procedure is complete, a window opens containing the message, **The server data are generated successfully**.
2. Perform this procedure for each primary OS server.

4.2.6 Lifebeat Monitoring Setup

Note

Important

Only one single manual procedure is required for integrating the symbols of the PCS 7 option Lifebeat Monitoring in SIMATIC PCS 7. This step only has to be performed once on the system of your first server project and not repeated again subsequently.

1. Open the OS server project in SIMATIC Manager. In the component view right-click **OS Master Server** and select **Open Object**. Your project is opened in **WinCC Explorer**.
2. In the left section of the **WinCC Explorer** window click on the **Graphics Designer** icon. In the right section of the window a list with picture files (with the file name extension .PDL) is displayed. In the right pane right-click on the file **@@PowerControlConfigTypicals.PDL**. Select **Open Picture** in the drop-down menu. The Graphics Designer opens the picture for editing.
3. Select all the symbols in the file **@@PowerControlConfigTypicals.pdl**. Select the following in the menu bar of the program:
Edit > Select All
Edit > Copy
4. Close the graphic. In response to the question whether the changes are to be saved, select **No**. The file is closed.
5. Open the **@@ConfigTypicals.pdl** file. Insert the PowerControl symbols into the file **@@ConfigTypicals.pdl** with **Paste**. To do this click on the empty space on the right next to the respective PCS 7 symbol and select **Edit > Paste**. While the symbols continue to be displayed as selected, move the entire selection to the right, if necessary, so that these can be displayed apart from the original PCS 7 symbols.
6. In order for all future projects to contain the right typicals for the Lifebeat Monitoring with PowerControl, copy the file created in step 5 into the following directory: **ProgramFiles \SIEMENS\WinCC\options\pd\base_Data_Pool\@@ConfigTypicals.PDL**.

Note

If you do not perform this step you have to repeat this procedure for each additional server project created in the future.

7. Select **File > Close** and then **Save**. A **Performance Alarm** dialog box opens. The alarm message says the file contains cyclical triggers. It is safe to ignore this warning. Click **OK**. Close the Graphics Designer.

4.2.7 Configure Alarm Logging

For further information, refer to the help file **Start > SIMATIC > WinCC > WinCC Information System > Working with WinCC > Setting Up a Message System**.

4.2.8 Configure Tag Logging

For further information, refer to the help file **Start > SIMATIC > WinCC > WinCC Information System > Working with WinCC > Archiving Process Values**.

Note

When using the Tag Logging Wizard, it is recommended that you name the archive **SystemArchive** for compatibility with PCS 7 trending faceplates.

4.3 Client Project Setup

For every client in the system, the following configuration steps are required:

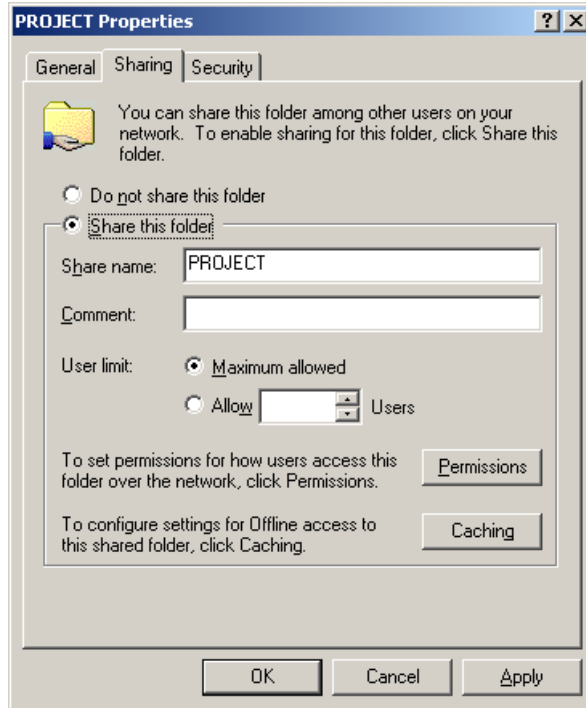
Step	Section
Create Shared Folders (Page 52)	4.3.1
Configure Client Station (Page 53)	4.3.2
Assign OS Server (Page 55)	4.3.3
Configure a Standard Server (Page 56)	4.3.4
Configure a Preferred Server (Page 56)	4.3.5
Run OS Project Editor (optional) (Page 56)	4.3.6
Lifebeat Monitoring Setup (Page 58)	4.3.7

4.3.1 Create Shared Folders

Shared folders on the clients are needed to store the project created on the Engineering Station. When the project is downloaded, it will be transferred from the Engineering Station to these folders. You cannot configure your stations in SIMATIC Manager until these folders are created and properly set up. Follow these steps to create and set up these folders, if they don't already exist.

Shared folders must be created for all Clients. Follow the procedure described in Section 3.2.1 (Page 43) to create a shared folder.

1. Using Windows Explorer, create a new folder such as "PROJECT" on one of the drives of the Primary and Standby servers, as well as all clients. The client and server projects created on the Engineering Station shall be downloaded to this folder on each node. If desired, you may create a "Projects" folder under `\Program Files\Siemens\Step7\proj\` for your projects. However, the name of the folder and where you place it is up to you.
2. Set up sharing on the newly created PROJECT folder. Right-click the **PROJECT** folder and select **Sharing...** The **PROJECT Properties** dialog box opens.



3. Select the **Sharing** tab, select the **Share this folder** radio button.
4. Click the **Permissions** button. The **Permissions** window opens. Select **Everyone**, and then click the **Full Control** check box in the **Allow** column.
5. Click **OK** to close the **Permissions** window.
6. Click the **OK** button to close the **Project Properties** dialog box.

4.3.2 Configure Client Station

Identify the path to which the project will be downloaded on the Client.

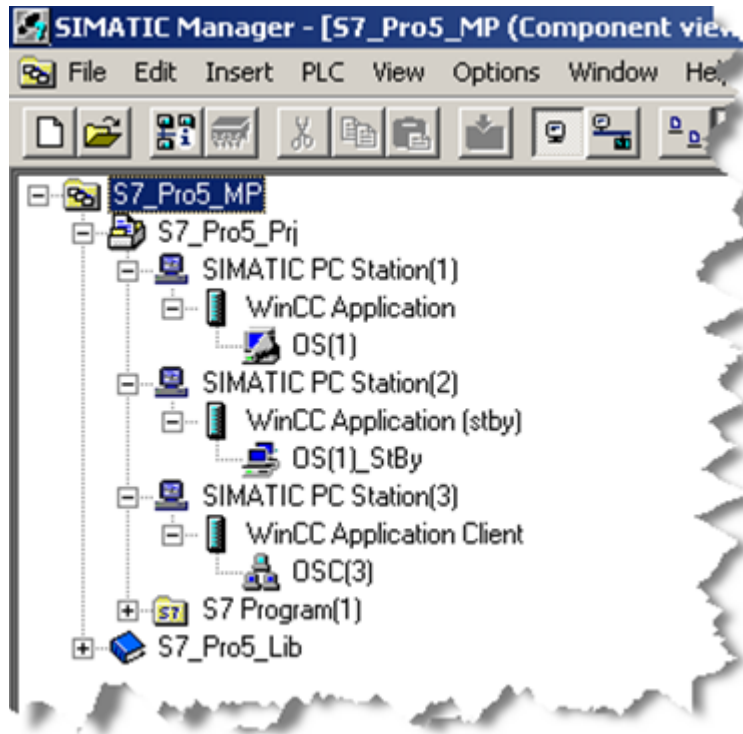
Note

If your client is not the same computer as your OS Engineering Station, then perform the following steps.

4.3 Client Project Setup

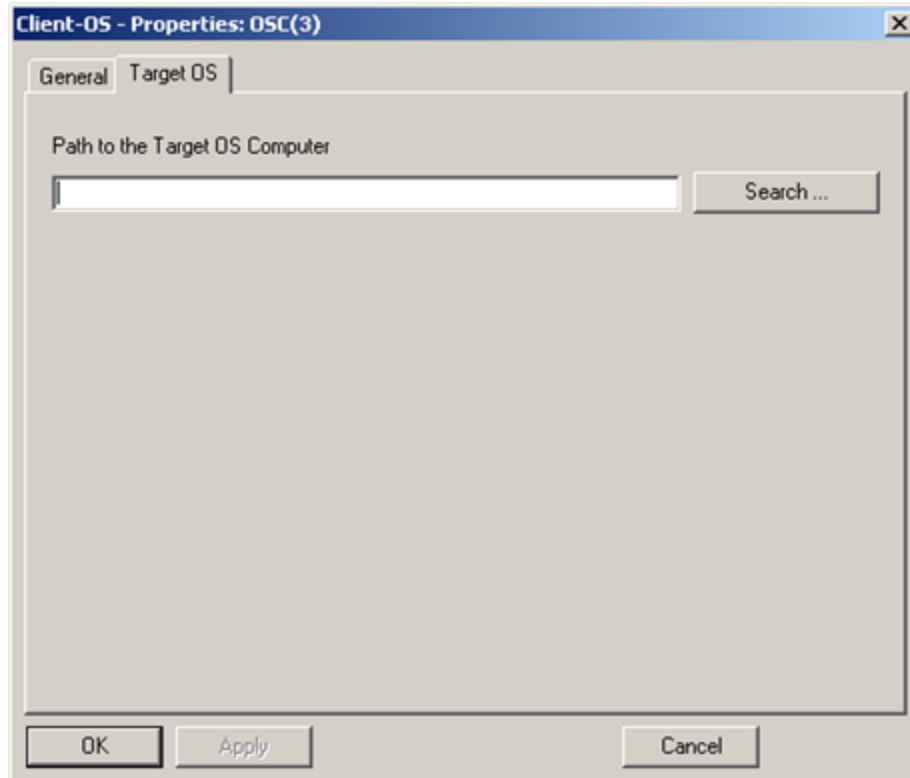
Follow these steps:

1. In the Component View of SIMATIC Manager, right-click the PC Station that is the Client to be configured and select **Object Properties**. For example, if your Client Server PC Stations is SIMATIC PC Station (3) in the figure below, right-click on **SIMATIC PC Station(3)**.



2. In the **General** tab, enter the computer name of the client computer in both the Name and Computer name fields. These names must be entered in ALL CAPS. Click the **OK** button.
3. Right-click **OSC(3)**, and select **Object Properties**.
The Properties dialog box opens.
4. Rename, as desired, the OSC(3) to the client project name.

5. Click the **Target OS** tab.



6. Select the Network button to browse and map to the shared project folder on OS Server. If a mapped drive already exists, select it from the available drives listed in the **Load In** list box. This is the path to the folder on your primary server where you will store and operate the project. This path must include the network name of the target computer.
Note: This folder must already exist on the primary and standby servers and be shared. This folder's permissions should be set to **Full Control** for **Everyone**. Refer to section **Create New Folders on the Primary and Standby Servers and All Clients** in this document. Click the **OK** button.
7. From the Component view, add and configure additional clients, servers, and server pairs as needed. You can add up to a total of 32 clients and 12 server pairs.

4.3.3 Assign OS Server

1. From the Component view in SIMATIC Manager, right-click OS Client and select **Assign OS server...** The OS server assignment window opens.
2. Check the project check box for each server as needed, and click the **OK** button. A progress bar indicating the status opens. When the procedure is complete, a window opens with the message, **The Procedure was complete without error**.
3. Click the **OK** button.

4.3.4 Configure a Standard Server

Note

The "Standard Server" represents the Server or Server pair that will store all operator actions and Runtime Trends. If a Standard Server is not configured, operator actions will be stored on the local machine instead of on the "Server". In addition, any Runtime Trends created will only be viewable from the client on which they are created.

On the OS Engineering Station, open the OS client project from SIMATIC Manager. Set up a standard server for Alarms and Split Screen Manager (SSM).

1. Right-click on **Server Data**.
2. Select **Standard Server**.
3. Click on the **Symb. Computer Name** field next to the **Alarms** Component. From the drop-down list, select the package that is the server for this client and is providing alarms for the client you are currently configuring.
4. Repeat step 3 for the **SSM** component.
5. Click the **OK** button.

This step can also be performed on the client after downloading the project to redefine the standard server for a particular client station. This customization will be overwritten during a subsequent download.

4.3.5 Configure a Preferred Server

For further information, please refer to the help file **Start > SIMATIC > WinCC > WinCC Information System > Options > Options for Process Control > Preferred Server**.

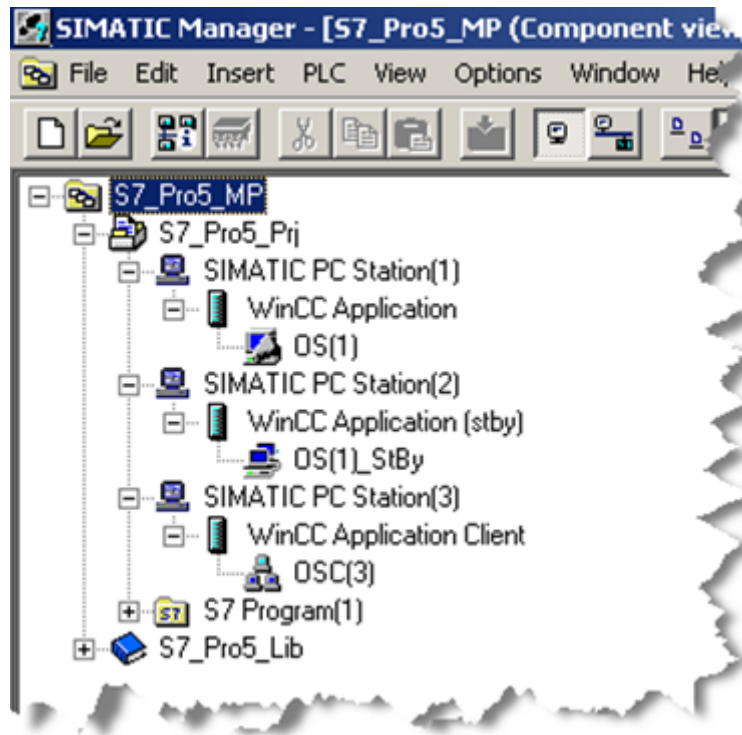
This step can also be performed on the client after downloading the project to customize the preferred server connection for a particular client station. This customization will be overwritten during a subsequent download.

4.3.6 Run OS Project Editor (optional)

If you want to change your layout, screen resolution, monitor configuration, message configuration, message display, area, runtime window, or basic data, run the OS project editor from WinCC Explorer.

To run the OS Project Editor:

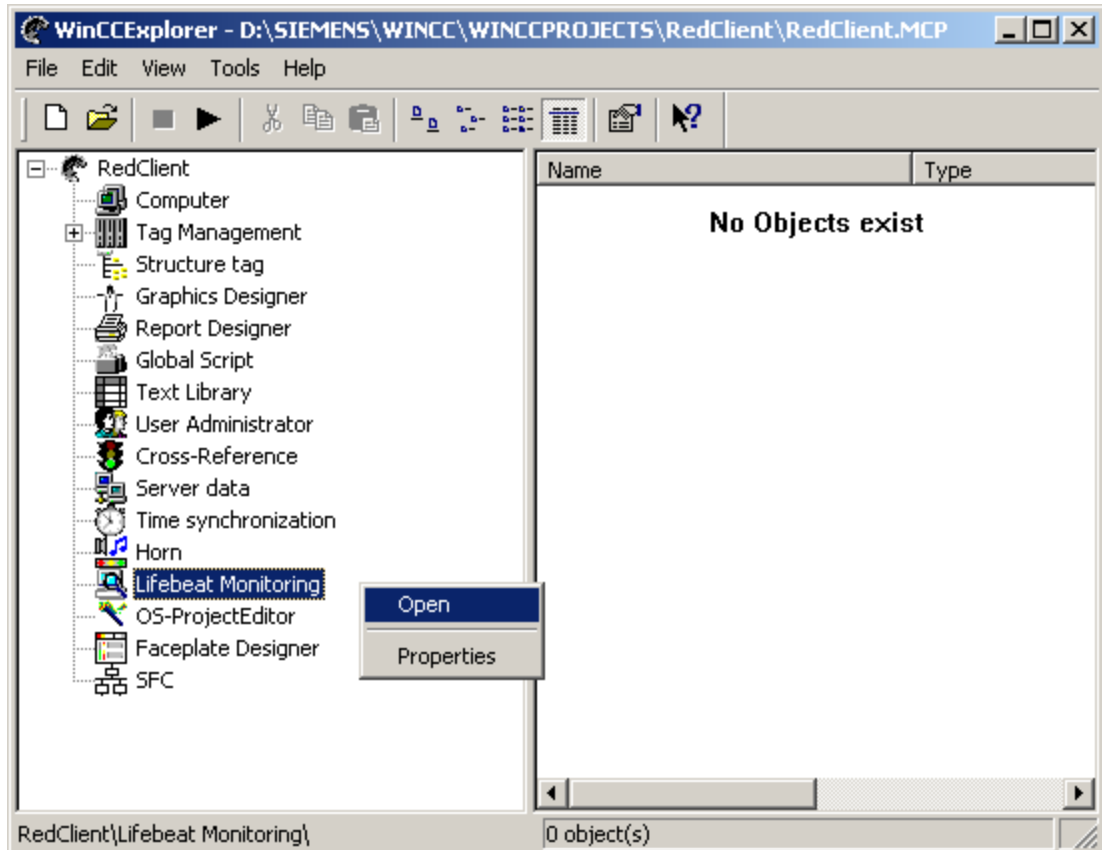
1. In the Component view of SIMATIC Manager, right-click the WinCC OS application and select **Open Object**. For example, if you want to run the OS Project Editor on the application OSC(3) in the figure below, right-click on **OSC(3)**. This will open the application using WinCC Explorer.



2. Select the OS Project Editor and make any desired changes. Refer to online help as you are running the OS Project Editor for more details.

4.3.7 Lifebeat Monitoring Setup

1. Open the client project. From the Component view in SIMATIC Manager, right-click on the OS client project and select **Open Object**.
The project opens in WinCC Explorer.



2. Right-click **Lifebeat Monitoring** and select **Open**.
The **Lifebeat Monitoring** window is displayed.

Lifebeat Monitoring - [RedClient.MCP]

Device List:

...	Device Name	Device Type	Connection
1	MRC_SERVERA	OS-Status	

System Configuration:

Device positioning in the system configuration

☒ automatic ☐ user-defined

Monitoring Cycle

☐ 5 s ☒ 10 s ☐ 1 min

Close

Update

Add

Delete

Print

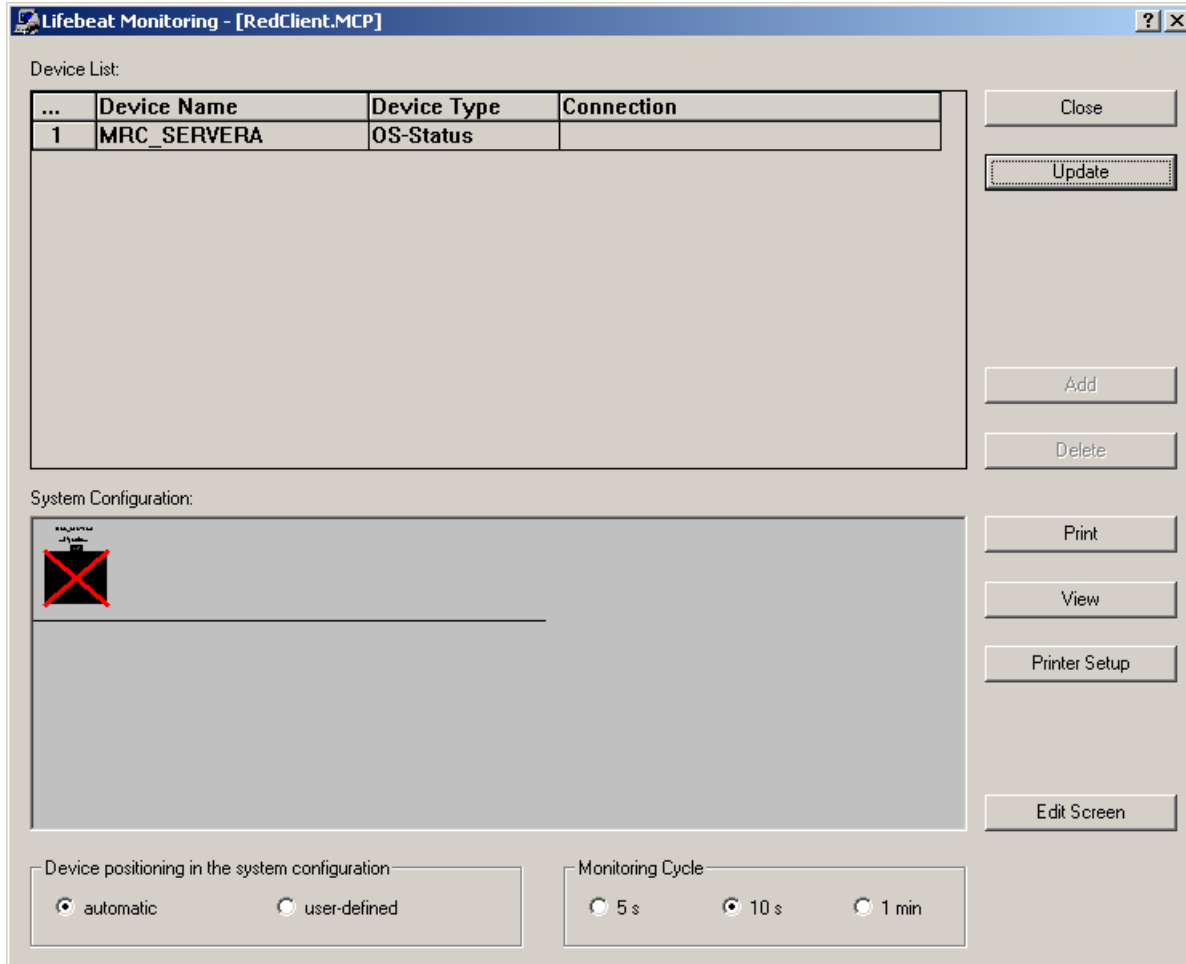
View

Printer Setup

Edit Screen

4.3 Client Project Setup

3. Set the Device positioning in the system configuration to automatic. Select a Monitoring Cycle time, and then click the **Update** button.
When complete, the system configuration section will update. See the Lifebeat Monitoring window below.



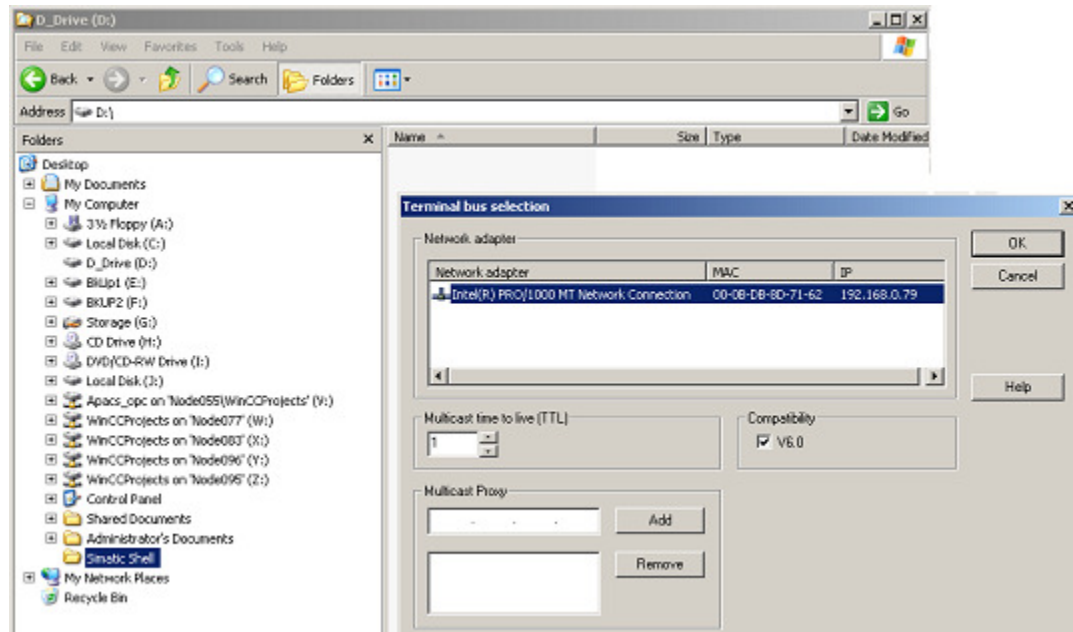
4. Lifebeat Monitor setup on the client is complete. Click the **Close** button to close Lifebeat Monitoring.

4.4 Configure the Terminal Bus

For Engineering Station, Central Archive Server and Client and Server PCs:

If the PC (computer) has more than one network adapter, configure WinCC for the network adapter it will be using.

1. Open Windows Explorer and select **My Computer > SIMATIC Shell > Settings**.
The **Terminal bus selection** window opens.



2. Select the appropriate network adapter. Click **OK**.
The message box **Network communications need to be initialized** opens.
3. Click **Yes**.

WinCC Communications Configurator

The WinCC Communications Configurator configures WinCC to use different network speeds (e.g. for dial-up or radio)

1. To start, click **Start > Simatic > WinCC > Tools > Communication Configurator**.
 - Slide bar should be set to reflect network speed.
 - "Server Pings Client" should be set unless there are dial-up clients.
 - Settings should only be changed if CCAgent, CCClient, and CCServer processes have been (manually) stopped.

Note

Refer to WinCC Help for additional details regarding WinCC Communication Configurator.

4.5 Time Synchronization Setup

For Clients and Server PCs:

For further information, please refer to the help file **Start > SIMATIC > WinCC > WinCC Information System > Options > Options for Process Control > Time Synchronization**.

4.6 User Administration

It is possible to configure all user accounts. For more details on configuring user accounts, refer to the help file:

Start>Simatic>WinCC>WinCC Information System>Working With WinCC>Setting up User Administration

4.7 Data Security and Backup

Recommendation

Save the various project states.

Create a backup in the following situations:

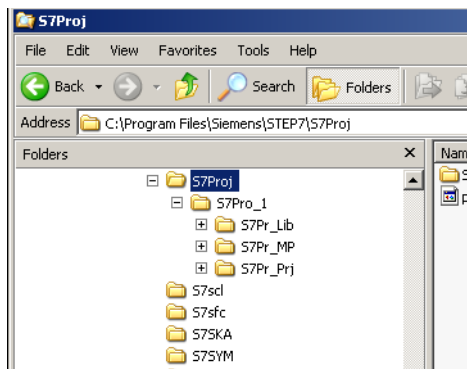
- After configuration changes
- Before and after system component upgrades
- Before and after the software update of the configuration software

Additional Information

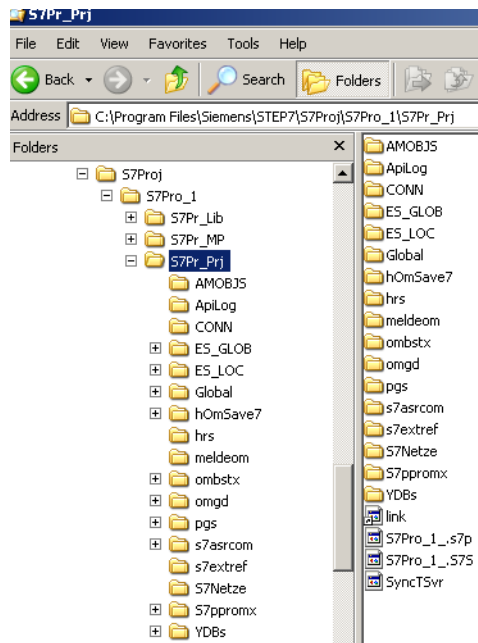
You will find a step-by-step description of saving and backing up ES and OS project data in the manual *Process Control System PCS 7; Service Support and Diagnostics*.

Important

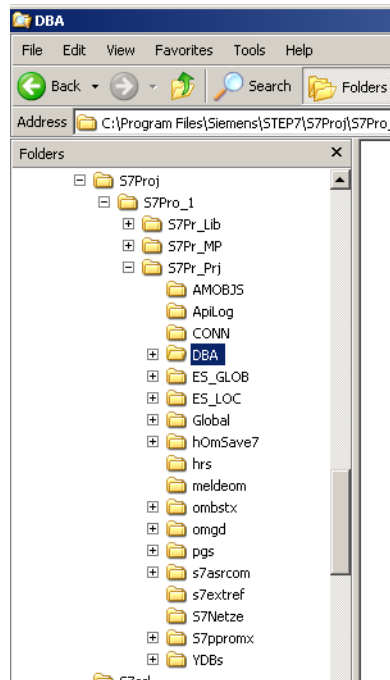
To insure that your DBA project is properly backed up using the standard steps for saving and backup up an ES project, it is recommended that your DBA project be stored in a folder within your SIMATIC Manager project folder. For example, if you create a SIMATIC Manager project called S7Pro_1, a folder hierarchy is automatically created:



Expanding the S7Pr_Prj directory shows many additional subfolders that are created automatically.



Create a new folder called DBA in the S7Pr_Prj subdirectory:



Your DBA project should be created in this subdirectory. This ensures that the DBA project will be automatically backed up and restored using the standard backup and restore procedures described in *Process Control System PCS 7; Service Support and Diagnostics*.

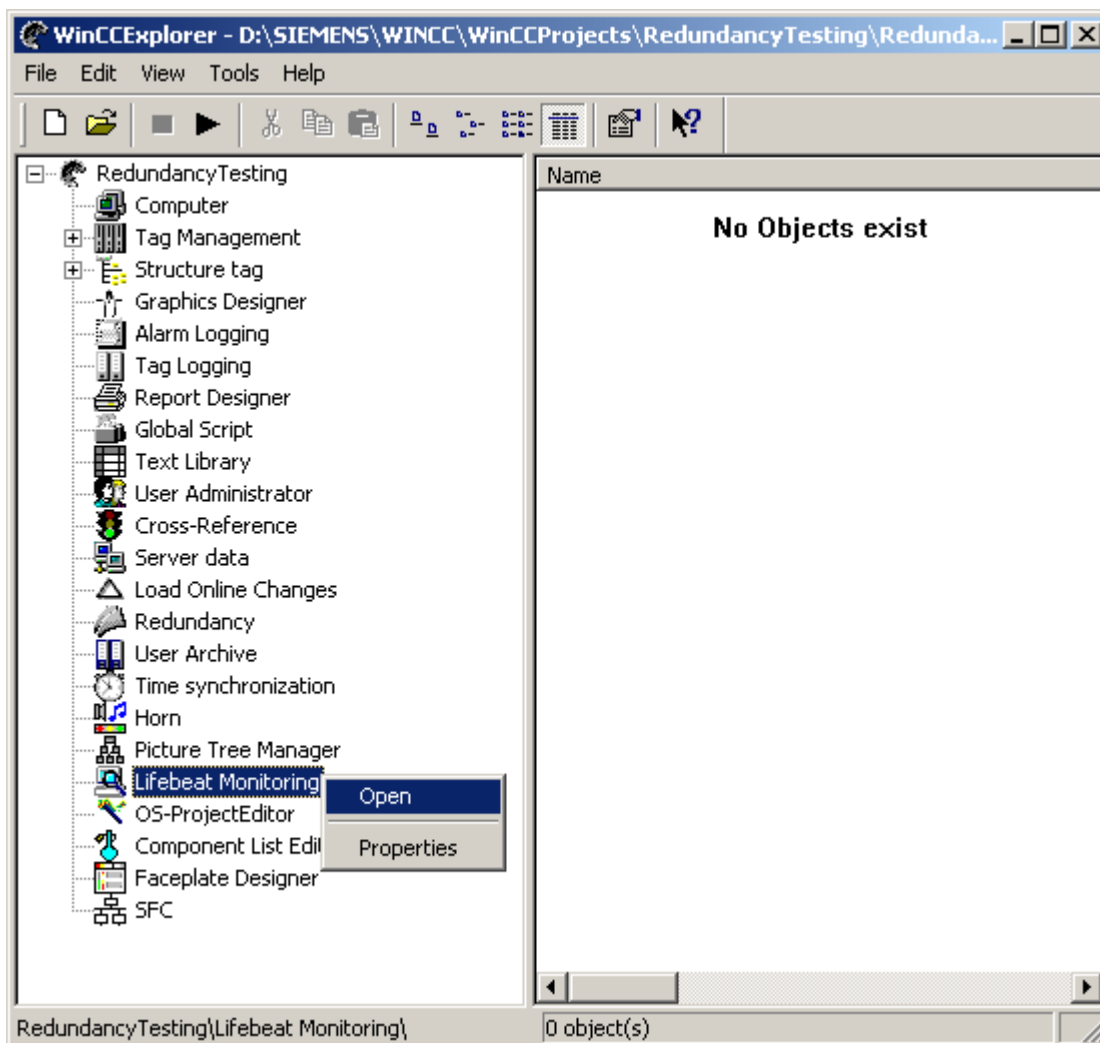
Note

It is highly recommended to backup user projects on a separate disk drive or partition.

Station Lifebeat Monitoring Setup

Once you have used DBA to create your PCS 7 PowerControl database(s), you can configure a Lifebeat Monitoring typical for each IED station and connecting device in the system. By following the simple procedure described below, a single graphic display is created automatically that provides a thumbnail representation of the connectivity of every hardware station in the system.

1. Open the OS server project in SIMATIC Manager. From the Component view, right-click on the **Primary OS Server** and select **Open Object**. Your project opens in **WinCC Explorer**.
2. In the left pane of WinCC Explorer, right-click **Lifebeat Monitoring**. Click **Open**.



The **Lifebeat Monitoring** window opens.

Lifebeat Monitoring - [RedundancyTesting.MCP]

?

×

Device List:

...	Device Name	Device Type	Connection
▶ 1			
2			
3			
4			
5			

System Configuration:

Device positioning in the system configuration

☒ automatic
 ☐ user-defined

Monitoring Cycle

☐ 5 s
 ☒ 10 s
 ☐ 1 min

Close

Update

Add

Delete

Print

View

Printer Setup

Edit Screen

3. In row 1, double-click the cell under **Device Name**. Enter the name of the station that you want to be monitored, make sure this is the name of the station defined in your DBA project.

Lifebeat Monitoring - [TELE_712_Release.mcp]

Device List:

...	Device Name	Device Type	Connection
▶ 1	IED001		
2			
3			
4			
5			
6			

System Configuration:

Device positioning in the system configuration:

☒ automatic ☐ user-defined

Monitoring Cycle:

☐ 5 s ☒ 10 s ☐ 1 min

Priority: 0

Buttons: Close, Update, Add, Delete, Print, Preview, Printer Setup, Edit Screen

4. Double-click the cell under Device Type, and select IED – IEC850 IED from the dropdown list.

The screenshot shows the 'Lifebeat Monitoring - [TELE_712_Release.mcp]' window. It contains a 'Device List' section with a table and a 'System Configuration' section with various controls.

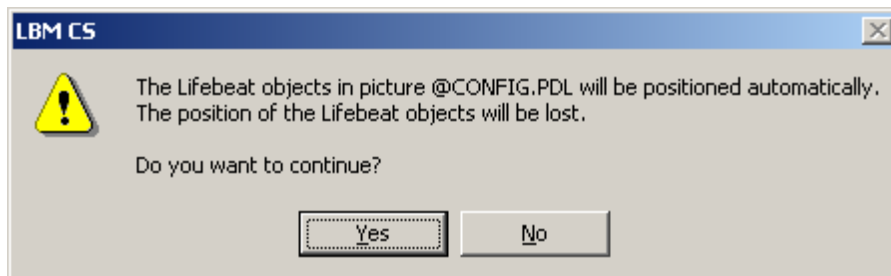
...	Device Name	Device Type	Connection
▶ 1	IED001		
2			
3		AS	
4		AS-2xx	
5		AS-3xx	
6		AS-4xx	
		Client	
		IEC850IED	
		OS	
		OS/Client	
		OS-OP	
		OS-PC	

Buttons on the right side of the 'Device List' section: Close, Update, Add, Delete, Print, Preview, Printer Setup, Edit Screen.

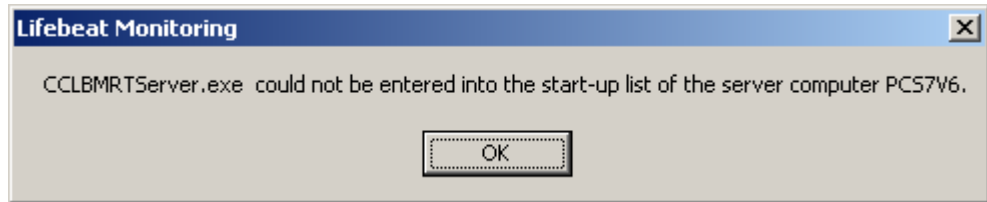
The 'System Configuration' section contains:

- Device positioning in the system configuration:
 - ☒ automatic
 - ☐ user-defined
- Monitoring Cycle:
 - ☐ 5 s
 - ☒ 10 s
 - ☐ 1 min
- Priority:

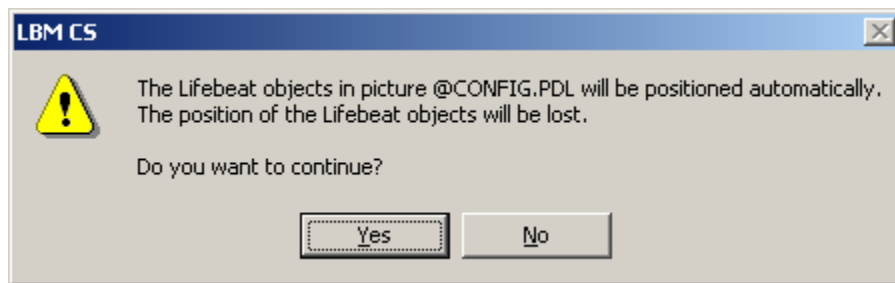
5. Repeat steps 2-4 for each station that you want monitored.
6. Under the Device positioning in the system configuration section, select **automatic**.
7. Select a **Monitoring Cycle time**. The default is 10 seconds.
8. When complete, select the **Update** button. The following dialog box is displayed.



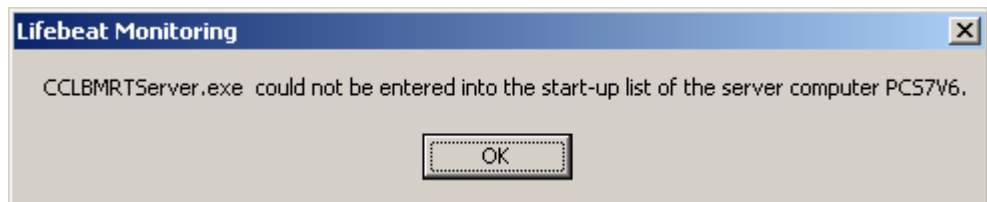
9. Click the **Yes** button. When complete, the **Lifebeat Monitoring** dialog box may appear indicating that the CCLBMRTServer.exe could not be entered into the start-up list of the server computer <computername>.



10. This message can be safely ignored. Click the **OK** button to continue.
11. Click the **Close** button to close the Lifebeat Monitoring window.
12. The system may respond with the following message "The process picture has not been updated, do you want to update before closing?" Click **Yes**.
13. The following dialog box may be displayed. Click **Yes**.



14. The following dialog box is displayed. Click **OK**.



15. To configure Lifebeat Monitoring to view the health of other system resources such as computers, for example OS Servers and the Archive Server, refer to **Start > SIMATIC > WinCC > WinCC Information System > Options > Options for Process Control > Lifebeat Monitoring > How to configure lifebeat monitoring with OPC connection**.

Downloading and Activating PCS 7 OS Stations

Downloading

Procedures vary depending upon the target of the download.

Activating and Reactivating

Perform activation and reactivation in the prescribed order.

6.1 Download the Project to the Primary Server

1. From SIMATIC Manager component view, right-click the Primary Server OS (PC Station 1), and select **PLC > Download**.
2. Ensure that **The Entire WinCC Project** radio button is selected.
3. Click **OK** to download the project.
4. A success window is displayed. Click **OK** to acknowledge that the download has completed.

6.2 Download the Project to the Standby Server

1. From SIMATIC Manager component view, right-click the Standby Server OS (PC Station 2), and select **PLC > Download**.
2. Ensure that **The Entire WinCC Project** radio button is selected.
3. Click **OK** to download the project.
4. A success window is displayed. Click **OK** to acknowledge that the download has completed.

6.3 Download the Project to the Client Station

1. From SIMATIC Manager component view, right-click the Client Station OS (PC Station 3), and select **PLC > Download**.
2. Ensure that **The Entire WinCC Project** radio button is selected.
3. Click **OK** to download the project.
4. A success window is displayed. Click **OK** to acknowledge that the download has completed.

6.4 Activating Server Projects

Note

Projects can be automatically activated when the operating system starts using the AutoStart Utility. For more information, refer to the help file within WinCC Explorer, or Internet Link (<https://support.industry.siemens.com/cs/ww/en/view/19249315>).

1. Activate the project on the primary server.
2. Wait four minutes for the project to fully start.
3. Activate the project on the backup server.
4. Wait four minutes for the project to fully start.
5. Activate the project on the client.

6.5 Restarting an OS Server

Wait five minutes before reactivating the WinCC runtime of an OS Server associated with a redundant pair to ensure that all synchronization activities will start-up properly.

Delta Engineering

This section describes how to make changes to an existing project from the Engineering Station, and then downloading those changes into the servers and client. These changes may include:

- Making changes to an AS program
- Adding or removing a station
- Making changes to the plant view in DBA
- Making changes to the project in SIMATIC Manager
- Making changes to the project in WinCC

7.1 Overall Steps

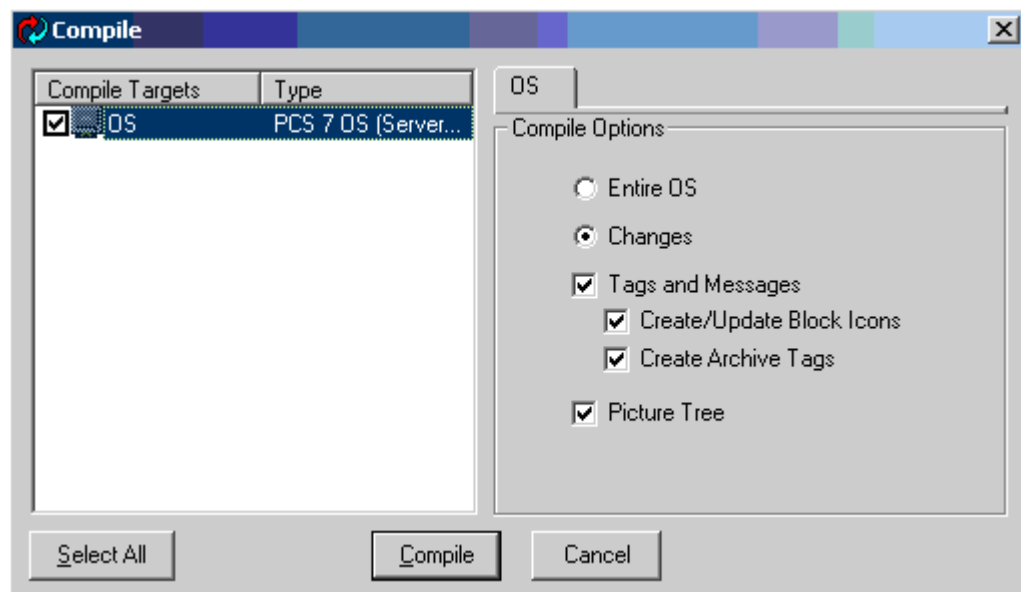
1. Make changes. Ensure that DBA changes have been compiled.
2. Download changes to your stations.

7.2 Making Changes to the AS Configuration

If necessary, use the appropriate Siemens or third-party tools to make changes to AS Station programs or hardware configuration.

7.3 Making Changes to the DBA Configuration

1. If you have made changes to your AS programs or hardware configuration, make any necessary corresponding changes to your DBA project (e.g. adding new object instances).
2. Use DBA to make any additional changes desired, such as
 - Fine-tuning of the Plant View assignments
 - Renaming folders or graphic files assigned to folders
 - Editing of AS Object attributes such as Symbol Names and Alarm Priorities (see DBA Help File, section Object Attributes, for more details)
3. From DBA, select Compile. If you have made changes to any DBA Types using the Type Editor, choose the "Entire OS" option. Otherwise, you may choose the "Changes" option on the **Compile** dialog box:



7.4 Modify the Project in WinCC

1. On the OS Engineering Station, start WinCC Explorer and open the OS Server project.
2. Make any new changes to the project that you need (for example, create new graphics, modify existing graphics, etc.).

Note

All changes must be made on the OS Engineering Station to preserve the **Changes Only** download. Changes made on another station cannot be downloaded to the other stations via SIMATIC Manager. These changes would have to be downloaded manually. Changes made on any station other than the Engineering Station will require a full download when future changes are made.

7.5 Download Changes

There are two methods to download changes:

- Load Online Changes (Delta Download)

Note**Important**

If you have removed objects from the plant view in DBA, have loaded online changes, and have later inserted the same objects back into the plant view in DBA, do not load these changes with the Downloading Changes Online function. For these objects to function properly in runtime, these changes must be loaded as part of a full downloading procedure.

- Full Download

Note

Certain changes made in DBA will require that the Full Compile method be used. You will be warned within DBA when attempting to make such changes that a Full Compile will be required.

7.5.1 Full Download

7.5.1.1 Download the Project to the Standby Server

1. From SIMATIC Manager component view, right-click the Standby Server OS (PC Station 2), and select **PLC > Download**.
2. Ensure that **The Entire WinCC Project** radio button is selected.
3. Click **OK** to download the project.
4. A success window is displayed. Click **OK** to acknowledge that the download has completed.

7.5.1.2 Download the Project to the Primary Server

1. From SIMATIC Manager component view, right-click the Primary Server OS (PC Station 1), and select **PLC > Download**.
2. Ensure that **The Entire WinCC Project** radio button is selected.
3. Click **OK** to download the project.
4. A success window is displayed. Click **OK** to acknowledge that the download has completed.

7.5.2 Downloading Changes

7.5.2.1 Download Change Overview

Load Online Changes (Delta Download)

1. From SIMATIC Manager component view, right-click the Primary Server OS (PC Station and select PLC > Download.
2. Ensure that "Changes" radio button is selected. (Note : In a redundant system, this choice is only available if both the Master and Standby are both online.
3. Click **OK** to download the changes. When downloading to redundant servers, the configured Standby will be downloaded first).
4. A success window is displayed. Click **OK** to acknowledge that the download has completed.
5. In a redundant system, the configured Master will then be downloaded and the OK button will again become available once this download is complete. Select **OK** to acknowledge the download.

For more details concerning Loading Online Changes, refer to SIMATIC Process Control System PCS 7 V9.1 Operator Station Configuration Manual.

Extending PCS 7 PowerControl

While PCS 7 PowerControl provides a set of typicals used to engineer a PowerControl system, it is possible to extend the system by creating your own typicals.

8.1 Creating New Technological Object Types

PCS 7 PowerControl DBA includes a powerful engineering environment, the DBA Type Editor, that allows user defined object types to be created. These can be created from "scratch", or extended based on the standard provided types. Refer to the *PCS 7 PowerControl DBA Type Editor* manual for complete details on how to create and modify object types.

8.2 Creating New Faceplates and Symbols

Note

For creating and modifying APL-style faceplates and symbols, refer to PCS 7 Libraries APL Style Guide.

When creating new object types, it is usually necessary to create a faceplate and one or more symbols that allow the HMI to interact with these objects.

Usually the best approach is to select a similar object from the standard library of objects and to "clone" the faceplate and symbol.

The following procedure describes this cloning process.

1. Select a standard faceplate that is most like the one you wish to create
Note: This exercise assumes you have selected the TC_COUNTER faceplate and symbol, and that you want to create a CUSTOM1_ faceplate and symbol from it.
2. If the new CUSTOM1_ faceplate is for a specific WinCC project, select the PDL files from the GraCS folder of the WinCC project.
If the modifications are to be global for all WinCC projects, select the files from the **SIEMENS\WINCC\options\pdl\FaceplateDesigner_V6** folder.
Any newly created WinCC project initially gets its files from the **SIEMENS\WINCC\options\pdl\FaceplateDesigner_V6** folder.

Creating New Faceplates

1. Using the WinCC Graphics Designer, open up the following PDL files, and use the **File > Save As..** menu option to save a copy of the file under a new name.

Original File Name	New Customized File Name
@PG_TC_COUNTER.pdl	@PG_CUSTOM1_.pdI
@PG_TC_COUNTER_OVERVIEW.pdl	@PG_CUSTOM1_OVERVIEW.pdl
@PG_TC_COUNTER_STANDARD.pdl	@PG_CUSTOM1_STANDARD.pdl
@PG_TC_COUNTER_VIEWLIST.pdl	@PG_CUSTOM1_VIEWLIST.pdl
@PL_TC_COUNTER.PDL	@PL_CUSTOM1_.PDL

1. All references to the old TC_COUNTER type need to be changed to reference the new CUSTOM1_ type. These changes are made in WinCC Graphics Designer by opening the PDL file, and modifying the reference using the object's property dialog box.
The following table lists all of the references that must be changed.

PDL FileName	Object Name	Property Name
@PG_TC_COUNTER.pdl	@Faceplate	FirstView
@PG_TC_COUNTER.pdl	ViewList	Picture Name
@PG_TC_COUNTER.pdl	BlockType	Output Value
@PL_TC_COUNTER.pdl	OverviewWindow	Picture Name
@PL_TC_COUNTER.pdl	View1	Picture Name
@PL_TC_COUNTER.pdl	BlockType	Output Value

The result is a copy of the TC_COUNTER faceplate that supports the new CUSTOM1_ type.

The objects on the new PDL files can now be added/deleted/modified. How to modify the PDL files is now up to the user, and is beyond the scope of this document.

Creating New Symbols

1. Using the WinCC Graphics Designer, open up the @@PowerControlTypicals.PDL file.
2. Save this under the new name @PowerControlTypicals.PDL.
3. Make a copy of the @ TC_COUNTER/2 object and name it @CUSTOM1_/1.
4. Using the property dialog box for the new @CUSTOM1_/1 object, change all references to the old TC_COUNTER type to reference the new CUSTOM1_ type.
5. The following properties must be changed:
 - Type
 - Server name

The result is a new symbol that will be used for the new CUSTOM1_ type.